

Question Bank

Complete Solutions

Web Technologies & Advanced JavaScript
jQuery · AJAX · PHP · MySQL

29 Questions · Ready-to-Run Code · Execution Instructions

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Set 1

jQuery, HTML/JavaScript & PHP/MySQL — Q1 to Q19

Q1. Dynamic To-Do List with jQuery

BROWSER

Problem Statement

Create a To-Do List where users can add new tasks, delete tasks via a Remove button, and toggle task completion (strike-through) on click. Use jQuery event delegation for dynamically added tasks. Apply `fadeIn()` when adding and `fadeOut()` when removing tasks.

File(s)

todo.html

Final Working Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Dynamic To-Do List</title>
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<style>
  body { font-family: Arial, sans-serif; max-width: 500px; margin: 40px auto; padding: 0 15px; }
  h2 { color: #333; }
  #taskInput { padding: 10px; width: 65%; border: 1px solid #ccc; border-radius: 4px; }
  #addBtn { padding: 10px 18px; background: #4CAF50; color: white; border: none; border-radius: 4px; cursor: pointer; }

  #addBtn:hover { background: #45a049; }
  .task { padding: 12px; margin: 8px 0; background: #f4f4f4; border-left: 4px solid #4CAF50;
    cursor: pointer; display: flex; justify-content: space-between; align-items: center; }
  .completed { text-decoration: line-through; opacity: 0.6; border-left-color: #999; }
  .remove { color: #e74c3c; font-weight: bold; cursor: pointer; padding: 0 8px; font-size: 18px; }
  .remove:hover { color: #c0392b; }
</style>
</head>
<body>
  <h2>My To-Do List</h2>
  <input type="text" id="taskInput" placeholder="Enter a new task...">
  <button id="addBtn">Add Task</button>
  <div id="taskList"></div>

<script>
$(document).ready(function () {
  // Add a new task
  $('#addBtn').click(function () {
    var taskText = $('#taskInput').val().trim();
    if (taskText === '') {
      alert('Please enter a task.');
```

```
    $(this).closest('.task').fadeOut(400, function () { $(this).remove(); });  
  });  
</script>  
</body>  
</html>
```

Execution Instructions

Save the code as todo.html anywhere on your computer. Double-click the file to open it in any browser (Chrome, Firefox, Edge). No server required.

Q2. Interactive Image Slider with Auto-Play

BROWSER

Problem Statement

Build an image slider that displays 3-5 images sequentially, auto-plays every 3 seconds using fadeIn/fadeOut, has Next and Previous buttons. Stop auto-play when the user hovers over the slider.

File(s)

slider.html

Final Working Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>jQuery Image Slider</title>
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<style>
body { font-family: Arial; text-align: center; background: #222; color: #fff; }
h2 { margin-top: 30px; }
.slider {
  position: relative; width: 600px; height: 350px; margin: 30px auto;
  overflow: hidden; border: 4px solid #fff; border-radius: 8px;
}
.slide {
  position: absolute; top: 0; left: 0; width: 100%; height: 100%;
  display: none;
}
.slide img { width: 100%; height: 100%; object-fit: cover; }
.nav-btn {
  position: absolute; top: 50%; transform: translateY(-50%);
  background: rgba(0,0,0,0.6); color: white; border: none;
  padding: 12px 16px; cursor: pointer; font-size: 18px; border-radius: 4px;
}
.nav-btn:hover { background: rgba(0,0,0,0.9); }
#prev { left: 10px; }
#next { right: 10px; }
.indicator { margin-top: 10px; color: #ccc; }
</style>
</head>
<body>
<h2>Image Slider</h2>
<div class="slider" id="slider">
  <div class="slide"></div>
  <div class="slide"></div>
  <div class="slide"></div>
  <div class="slide"></div>
  <button id="prev" class="nav-btn">&#10094;</button>
  <button id="next" class="nav-btn">&#10095;</button>
</div>
<div class="indicator" id="indicator">Slide 1</div>

<script>
$(document).ready(function () {
  var $slides = $('<div>');
  var current = 0;
  var total = $slides.length;
  var autoPlay;

  function showSlide(index) {
    $slides.fadeOut(500);
    $slides.eq(index).fadeIn(500);
    $('#indicator').text('Slide ' + (index + 1) + ' of ' + total);
  }

  function nextSlide() {
    current = (current + 1) % total;
    showSlide(current);
  }
}
```

```

function prevSlide() {
    current = (current - 1 + total) % total;
    showSlide(current);
}

function startAutoPlay() { autoPlay = setInterval(nextSlide, 3000); }
function stopAutoPlay() { clearInterval(autoPlay); }

$('#next').click(function () { nextSlide(); });
$('#prev').click(function () { prevSlide(); });

// Stop auto-play on hover, resume on mouse leave
$('#slider').hover(stopAutoPlay, startAutoPlay);

showSlide(current);
startAutoPlay();
});
</script>
</body>
</html>

```

Execution Instructions

Save as slider.html and open in any browser. Uses public placeholder images from picsum.photos so no local image files are required.

Q3. Real-Time Username Availability Checker

BROWSER

Problem Statement

Develop a signup form where as the user types a username, an AJAX request checks if it's available. Display 'Available' (green) or 'Taken' (red) without page reload. Use jQuery \$.post() (simulated) and slideDown() to show the status message.

File(s)

username_check.html

Final Working Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Username Availability Checker</title>
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<style>
  body { font-family: Arial; max-width: 420px; margin: 60px auto; padding: 20px;
    background: #f9f9f9; border: 1px solid #ddd; border-radius: 6px; }
  h2 { color: #333; }
  input { width: 100%; padding: 10px; box-sizing: border-box; font-size: 14px;
    border: 1px solid #ccc; border-radius: 4px; }
  #status { display: none; margin-top: 10px; padding: 10px; border-radius: 4px;
    font-weight: bold; }
  .available { background: #d4edda; color: #155724; }
  .taken { background: #f8d7da; color: #721c24; }
  .checking { background: #fff3cd; color: #856404; }
</style>
</head>
<body>
  <h2>Sign Up</h2>
  <label>Username:</label>
  <input type="text" id="username" placeholder="Type a username...">
  <div id="status"></div>

<script>
$(document).ready(function () {
  // Simulated server: list of "taken" usernames
  var takenUsernames = ['admin', 'john', 'test', 'user', 'sann', 'mary'];

  // Override AJAX with a simulated response so it works without a backend.
  // In a real app you would call $.post('check_username.php', ...) instead.
  function checkUsername(name, callback) {
    setTimeout(function () {
      var isTaken = takenUsernames.indexOf(name.toLowerCase()) !== -1;
      callback({ available: !isTaken });
    }, 400); // simulate network delay
  }

  var typingTimer;
  $('#username').on('keyup', function () {
    clearTimeout(typingTimer);
    var name = $(this).val().trim();
    var $status = $('#status');

    if (name.length < 3) {
      $status.slideUp();
      return;
    }

    $status.removeClass('available taken').addClass('checking')
      .text('Checking...').slideDown();

    // Wait 300ms after the user stops typing, then "POST" the check
    typingTimer = setTimeout(function () {
      // --- This is where $.post() would be used in production: ---
      // $.post('check_username.php', { username: name }, function (data) { ... });
    }, 300);
  });
});
```



```

    checkUsername(name, function (data) {
        $status.removeClass('checking');
        if (data.available) {
            $status.removeClass('taken').addClass('available')
                .text('"' + name + '" is available!');
        } else {
            $status.removeClass('available').addClass('taken')
                .text('"' + name + '" is already taken.');
```

```

        }
        $status.hide().slideDown();
    });
}, 300);
});
</script>
</body>
</html>
```

Execution Instructions

Save as username_check.html and open in any browser. The 'taken' usernames are simulated locally inside a fake \$.post() handler so it works without a backend. Try usernames like 'admin', 'john', or 'test' to see the 'taken' response.

Q4. Dynamic FAQ Accordion

BROWSER

Problem Statement

Create an FAQ section where clicking a question slides down its answer; clicking again slides up the answer (accordion effect). Use `slideToggle()` for smooth animations and highlight the active question with a different background.

File(s)

faq.html

Final Working Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>FAQ Accordion</title>
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<style>
  body { font-family: Arial; max-width: 700px; margin: 40px auto; padding: 0 20px; }
  h2 { text-align: center; color: #333; }
  .faq-item { border: 1px solid #ddd; margin: 8px 0; border-radius: 4px; overflow: hidden; }
  .faq-question {
    padding: 14px 18px; background: #f4f4f4; cursor: pointer;
    font-weight: bold; user-select: none; transition: background 0.3s;
  }
  .faq-question:hover { background: #ecec; }
  .faq-question.active { background: #4CAF50; color: white; }
  .faq-answer { display: none; padding: 14px 18px; background: white; line-height: 1.6; }
</style>
</head>
<body>
  <h2>Frequently Asked Questions</h2>

  <div class="faq-item">
    <div class="faq-question">What is jQuery?</div>
    <div class="faq-answer">jQuery is a fast, small, and feature-rich JavaScript library that
      simplifies HTML document traversal, event handling, animation, and AJAX.</div>
  </div>

  <div class="faq-item">
    <div class="faq-question">Is jQuery still relevant in 2026?</div>
    <div class="faq-answer">Yes. While modern frameworks like React and Vue dominate new
      projects, jQuery still powers a huge portion of existing websites and is great for
      quick DOM manipulation tasks.</div>
  </div>

  <div class="faq-item">
    <div class="faq-question">How do I include jQuery in my project?</div>
    <div class="faq-answer">Add the script tag pointing to a CDN such as
      https://code.jquery.com/jquery-3.6.0.min.js in your HTML head section.</div>
  </div>

  <div class="faq-item">
    <div class="faq-question">What is slideToggle() used for?</div>
    <div class="faq-answer">slideToggle() animates an element between hidden and visible
      states using a sliding effect, perfect for accordions and dropdowns.</div>
  </div>

</body>
</html>

<script>
$(document).ready(function () {
  $('.faq-question').click(function () {
    var $this = $(this);
    var $answer = $this.next('.faq-answer');

    // Close all other open answers (accordion behavior)
    $('.faq-answer').not($answer).slideUp(300);
    $('.faq-question').not($this).removeClass('active');
  });
});
```

```
// Toggle the clicked one
$answer.slideToggle(300);
$this.toggleClass('active');
});
</script>
</body>
</html>
```

Execution Instructions

Save as `faq.html` and open in any browser.

Q5. AJAX-Based Live Search Filter

BROWSER

Problem Statement

Implement a search bar that filters a list of products as the user types, using jQuery \$.get() to fetch matching results dynamically. Show 'No results found' if no matches exist. Use fadeIn() to display filtered results smoothly.

File(s)

live_search.html

Final Working Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Live Search Filter</title>
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<style>
  body { font-family: Arial; max-width: 500px; margin: 40px auto; padding: 0 20px; }
  h2 { color: #333; }
  #search { width: 100%; padding: 12px; box-sizing: border-box; font-size: 15px;
    border: 1px solid #ccc; border-radius: 4px; }
  #results { margin-top: 10px; }
  .product { display: none; padding: 12px; background: #f4f4f4; margin: 6px 0;
    border-radius: 4px; border-left: 4px solid #2196F3; }
  .no-results { padding: 12px; color: #c0392b; font-style: italic; }
</style>
</head>
<body>
  <h2>Product Search</h2>
  <input type="text" id="search" placeholder="Search for products...">
  <div id="results"></div>

<script>
$(document).ready(function () {
  // Mock data that would normally come from $.get('products.json', ...)
  var allProducts = [
    { name: 'iPhone 15 Pro', category: 'Mobiles' },
    { name: 'Samsung Galaxy S24', category: 'Mobiles' },
    { name: 'OnePlus 12', category: 'Mobiles' },
    { name: 'Dell XPS 15', category: 'Laptops' },
    { name: 'MacBook Pro M3', category: 'Laptops' },
    { name: 'HP Spectre x360', category: 'Laptops' },
    { name: 'Sony WH-1000XM5', category: 'Headphones' },
    { name: 'AirPods Pro', category: 'Headphones' },
    { name: 'iPad Air', category: 'Tablets' },
    { name: 'Samsung Tab S9', category: 'Tablets' }
  ];

  // Simulate $.get() returning JSON. In a real app:
  // $.get('products.json', function(data) { ... });
  function fetchProducts(query, callback) {
    setTimeout(function () {
      var filtered = allProducts.filter(function (p) {
        return p.name.toLowerCase().indexOf(query.toLowerCase()) !== -1 ||
          p.category.toLowerCase().indexOf(query.toLowerCase()) !== -1;
      });
      callback(filtered);
    }, 150);
  }

  $('#search').on('keyup', function () {
    var query = $(this).val().trim();
    var $results = $('#results');
    $results.empty();

    if (query === '') return;

    fetchProducts(query, function (data) {
      if (data.length === 0) {

```

```

    $results.html('<div class="no-results">No results found for "' + query + '"</div>');
  } else {
    data.forEach(function (p) {
      var $item = $('<div class="product"></div>')
        .html('<strong>' + p.name + '</strong> <em>(' + p.category + ')</em>');
      $results.append($item);
      $item.fadeIn(300);
    });
  }
});
});
</script>
</body>
</html>

```

Execution Instructions

Save as live_search.html and open in any browser. The product list is included as a JS array; the structure mimics what a \$.get() call returning JSON would look like.

Q6. Drag-and-Drop Shopping Cart

BROWSER

Problem Statement

Build a product listing page where users can drag items into a shopping cart and see the cart update dynamically. Use jQuery UI draggable/droppable. Animate cart updates with slideDown().

File(s)

drag_cart.html

Final Working Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Drag & Drop Shopping Cart</title>
<link rel="stylesheet" href="https://code.jquery.com/ui/1.13.2/themes/base/jquery-ui.css">
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<script src="https://code.jquery.com/ui/1.13.2/jquery-ui.min.js"></script>
<style>
  body { font-family: Arial; margin: 30px; }
  h2 { color: #333; }
  .container { display: flex; gap: 30px; }
  .products, .cart {
    width: 45%; min-height: 400px; padding: 15px;
    border: 2px dashed #aaa; border-radius: 6px;
  }
  .cart { border-color: #4CAF50; background: #f0fff4; }
  .product {
    padding: 12px; margin: 8px 0; background: #2196F3; color: white;
    border-radius: 4px; cursor: grab; user-select: none;
  }
  .product:active { cursor: grabbing; }
  .cart-item {
    display: none; padding: 10px; margin: 6px 0; background: #fff;
    border-left: 4px solid #4CAF50; border-radius: 4px;
  }
  #total { margin-top: 15px; font-weight: bold; font-size: 18px; }
</style>
</head>
<body>
  <h2>Shop &mdash; drag products into the cart</h2>
  <div class="container">
    <div class="products">
      <h3>Available Products</h3>
      <div class="product" data-name="Headphones" data-price="49.99">Headphones - $49.99</div>
      <div class="product" data-name="Keyboard" data-price="29.99">Keyboard - $29.99</div>
      <div class="product" data-name="Mouse" data-price="19.99">Mouse - $19.99</div>
      <div class="product" data-name="Monitor" data-price="199.99">Monitor - $199.99</div>
      <div class="product" data-name="Webcam" data-price="59.99">Webcam - $59.99</div>
    </div>
    <div class="cart" id="cart">
      <h3>Shopping Cart</h3>
      <div id="cartItems"></div>
      <div id="total">Total: $0.00</div>
    </div>
  </div>

  <script>
$(document).ready(function () {
  var total = 0;

  $('.product').draggable({
    helper: 'clone',
    revert: 'invalid'
  });

  $('#cart').droppable({
    accept: '.product',
    drop: function (event, ui) {
```

```

    var name = ui.draggable.data('name');
    var price = parseFloat(ui.draggable.data('price'));

    var $item = $('<div class="cart-item"></div>')
        .text(name + ' - $' + price.toFixed(2));
    $('#cartItems').append($item);
    $item.slideDown(300);

    total += price;
    $('#total').text('Total: $' + total.toFixed(2));
  }
});
</script>
</body>
</html>

```

Execution Instructions

Save as drag_cart.html and open in any browser. Uses jQuery UI from CDN. Drag any product into the cart on the right.

Q7. Dynamic Progress Bar for File Upload

BROWSER

Problem Statement

Simulate a file upload system where a progress bar fills up over 5 seconds when 'Upload' is clicked, then shows 'Upload Complete!' at the end. Use jQuery animate() for the progress bar and fadeIn() for the success message.

File(s)

progress.html

Final Working Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>File Upload Progress Bar</title>
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<style>
  body { font-family: Arial; max-width: 500px; margin: 60px auto; padding: 0 20px;
    text-align: center; }
  h2 { color: #333; }
  #uploadBtn { padding: 12px 24px; background: #2196F3; color: white;
    border: none; border-radius: 4px; font-size: 15px; cursor: pointer; }
  #uploadBtn:hover { background: #1976D2; }
  #uploadBtn:disabled { background: #aaa; cursor: not-allowed; }
  .bar-wrap { background: #eee; border-radius: 12px; height: 24px;
    margin: 25px 0 10px; overflow: hidden; }
  .bar { background: linear-gradient(90deg, #4CAF50, #8BC34A);
    height: 100%; width: 0%; color: white; line-height: 24px;
    text-align: center; font-weight: bold; font-size: 13px; }
  #message { display: none; margin-top: 15px; padding: 12px; background: #d4edda;
    color: #155724; border-radius: 4px; font-weight: bold; }
</style>
</head>
<body>
  <h2>File Upload Simulator</h2>
  <button id="uploadBtn">Upload</button>

  <div class="bar-wrap"><div class="bar" id="bar">0%</div></div>
  <div id="message">Upload Complete!</div>

<script>
$(document).ready(function () {
  $('#uploadBtn').click(function () {
    var $btn = $(this);
    var $bar = $('#bar');
    $btn.prop('disabled', true);
    $('#message').hide();
    $bar.css('width', '0%').text('0%');

    // Update text label as the bar grows
    $({ p: 0 }).animate({ p: 100 }, {
      duration: 5000,
      step: function (now) {
        var pct = Math.round(now);
        $bar.css('width', pct + '%').text(pct + '%');
      },
      complete: function () {
        $('#message').fadeIn(500);
        $btn.prop('disabled', false);
      }
    });
  });
});
</script>
</body>
</html>
```

Execution Instructions

Save as progress.html and open in any browser. Click the Upload button to start the simulated upload.

Q8. Interactive Quiz with Timer

BROWSER

Problem Statement

Create a quiz app where each question appears one by one, a 10-second timer counts down per question, and the final score is shown at the end. Use jQuery delay() and fadeOut() for transitions and update the score dynamically using text() or html().

File(s)

quiz.html

Final Working Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Quiz with Timer</title>
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<style>
  body { font-family: Arial; max-width: 600px; margin: 40px auto; padding: 0 20px;
    background: #f5f5f5; }
  .card { background: white; padding: 25px; border-radius: 8px;
    box-shadow: 0 2px 6px rgba(0,0,0,0.1); }
  h2 { color: #333; }
  #timer { float: right; font-weight: bold; color: #e74c3c; font-size: 18px; }
  .option { display: block; width: 100%; padding: 12px; margin: 8px 0;
    background: #f4f4f4; border: 1px solid #ddd; border-radius: 4px;
    cursor: pointer; text-align: left; font-size: 14px; }
  .option:hover { background: #e3f2fd; border-color: #2196f3; }
  #score { text-align: center; font-size: 22px; color: #4CAF50; }
</style>
</head>
<body>
  <div class="card">
    <span id="timer">10s</span>
    <h2 id="qNumber">Question 1</h2>
    <p id="question"></p>
    <div id="options"></div>
  </div>

<script>
$(document).ready(function () {
  var quiz = [
    { q: 'What does HTML stand for?',
      opts: ['Hyper Text Markup Language', 'High Tech Modern Language',
        'Hyperlink Machine Language', 'Home Tool Markup Language'],
      ans: 0 },
    { q: 'Which keyword declares a constant in JavaScript?',
      opts: ['var', 'let', 'const', 'static'], ans: 2 },
    { q: 'Which symbol is used for IDs in CSS?',
      opts: ['.', '#', '@', '*'], ans: 1 },
    { q: 'Which jQuery method shows an element gradually?',
      opts: ['show()', 'fadeIn()', 'appear()', 'visible()'], ans: 1 },
    { q: 'What does AJAX stand for?',
      opts: ['Asynchronous JavaScript and XML', 'Async Java And XML',
        'Auto JavaScript And XHTML', 'Advanced JS API XML'], ans: 0 }
  ];

  var index = 0, score = 0, timeLeft = 10, timerInt;

  function loadQuestion() {
    if (index >= quiz.length) { showFinal(); return; }
    var q = quiz[index];
    $('#qNumber').text('Question ' + (index + 1) + ' of ' + quiz.length);
    $('#question').text(q.q);
    $('#options').empty();
    q.opts.forEach(function (opt, i) {
      $('<button class="option"></button>')
        .text(opt).data('i', i).appendTo('#options');
    });
  }
});
```

```

});
timeLeft = 10;
$('#timer').text(timeLeft + 's');
clearInterval(timerInt);
timerInt = setInterval(tick, 1000);
}

function tick() {
    timeLeft--;
    $('#timer').text(timeLeft + 's');
    if (timeLeft <= 0) { clearInterval(timerInt); nextQuestion(); }
}

function nextQuestion() {
    $('#card').fadeOut(300).delay(100).fadeIn(300, function () {
        index++;
        loadQuestion();
    });
}

function showFinal() {
    clearInterval(timerInt);
    $('#card').html('<h2>Quiz Complete!</h2>' +
        '<p id="score">Your score: ' + score + ' / ' + quiz.length + '</p>');
}

$('#options').on('click', '.option', function () {
    if (parseInt($(this).data('i')) === quiz[index].ans) score++;
    clearInterval(timerInt);
    nextQuestion();
});

loadQuestion();
});
</script>
</body>
</html>

```

Execution Instructions

Save as quiz.html and open in any browser. Click an option to answer; if you don't answer in 10 seconds the quiz auto-advances.

Q9. Animated Login/Signup Form Toggle

BROWSER

Problem Statement

Design a login page where clicking 'Sign Up' slides in the registration form and clicking 'Login' slides back to the login form. Use `slideToggle()` for smooth transitions and validate inputs before submission.

File(s)

login_signup.html

Final Working Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Login / Signup Toggle</title>
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<style>
  body { font-family: Arial; background: #eef2f7; }
  .container { max-width: 380px; margin: 60px auto; background: white;
    padding: 25px; border-radius: 8px; box-shadow: 0 2px 8px rgba(0,0,0,0.1); }
  h2 { text-align: center; color: #333; margin-top: 0; }
  .toggle-links { text-align: center; margin-bottom: 15px; }
  .toggle-links a { margin: 0 10px; color: #2196F3; cursor: pointer; text-decoration: none;
    font-weight: bold; }
  .toggle-links a.active { text-decoration: underline; }
  input { width: 100%; padding: 10px; box-sizing: border-box;
    margin: 5px 0 12px; border: 1px solid #ccc; border-radius: 4px; }
  button { width: 100%; padding: 12px; background: #4CAF50; color: white;
    border: none; border-radius: 4px; cursor: pointer; font-size: 15px; }
  button:hover { background: #45a049; }
  .error { color: #c0392b; font-size: 13px; margin-bottom: 10px; }
  #signupForm { display: none; }
</style>
</head>
<body>
  <div class="container">
    <div class="toggle-links">
      <a id="loginLink" class="active">Login</a> |
      <a id="signupLink">Sign Up</a>
    </div>

    <form id="loginForm">
      <h2>Login</h2>
      <input type="email" id="loginEmail" placeholder="Email">
      <input type="password" id="loginPassword" placeholder="Password">
      <div class="error" id="loginError"></div>
      <button type="submit">Login</button>
    </form>

    <form id="signupForm">
      <h2>Sign Up</h2>
      <input type="text" id="signupName" placeholder="Full Name">
      <input type="email" id="signupEmail" placeholder="Email">
      <input type="password" id="signupPassword" placeholder="Password (min 6 chars)">
      <div class="error" id="signupError"></div>
      <button type="submit">Create Account</button>
    </form>
  </div>

  <script>
$(document).ready(function () {
  $('#signupLink').click(function () {
    $('#loginForm, #signupForm').slideToggle(400);
    $(this).addClass('active');
    $('#loginLink').removeClass('active');
  });
  $('#loginLink').click(function () {
    if ($('#signupForm').is(':visible')) {
      $('#loginForm, #signupForm').slideToggle(400);
    }
  });
});
</script>
```

```

        $(this).addClass('active');
        $('#signupLink').removeClass('active');
    }
});

$('#loginForm').submit(function (e) {
    e.preventDefault();
    var email = $('#loginEmail').val().trim();
    var pass = $('#loginPassword').val();
    if (!/^[\^s@]+\.[^\s@]+\.[^\s@]+$/i.test(email)) {
        $('#loginError').text('Please enter a valid email.').return;
    }
    if (pass.length < 6) {
        $('#loginError').text('Password must be at least 6 characters.').return;
    }
    $('#loginError').text('');
    alert('Login successful for ' + email);
});

$('#signupForm').submit(function (e) {
    e.preventDefault();
    var name = $('#signupName').val().trim();
    var email = $('#signupEmail').val().trim();
    var pass = $('#signupPassword').val();
    if (name === '') { $('#signupError').text('Name is required.').return; }
    if (!/^[\^s@]+\.[^\s@]+\.[^\s@]+$/i.test(email)) {
        $('#signupError').text('Please enter a valid email.').return;
    }
    if (pass.length < 6) {
        $('#signupError').text('Password must be at least 6 characters.').return;
    }
    $('#signupError').text('');
    alert('Account created for ' + name);
});
});
</script>
</body>
</html>

```

Execution Instructions

Save as login_signup.html and open in any browser. Click the toggle links above the form to switch between Login and Sign Up.

Q10. User Registration System

XAMPP REQUIRED

Problem Statement

Build a website registration system in PHP. Users sign up with name, email, password, and phone number. All fields must be validated (no empty inputs, valid email format, 10-digit phone number). Passwords must be hashed before storing in the MySQL database. Display error messages on validation failure. On successful registration, redirect to a welcome page.

File(s)

register.php (and welcome.php)

Final Working Code

```
-- =====
-- SQL: Run this once in phpMyAdmin BEFORE using the PHP file
-- =====
CREATE DATABASE IF NOT EXISTS qbank;
USE qbank;
CREATE TABLE IF NOT EXISTS users (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(150) NOT NULL UNIQUE,
    password VARCHAR(255) NOT NULL,
    phone VARCHAR(15) NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

<?php
// ===== FILE: register.php =====
$errors = [];
$name = $email = $phone = '';

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $name = trim($_POST['name'] ?? '');
    $email = trim($_POST['email'] ?? '');
    $password = $_POST['password'] ?? '';
    $phone = trim($_POST['phone'] ?? '');

    // Validation
    if ($name === '') $errors[] = 'Name is required.';
    if (!filter_var($email, FILTER_VALIDATE_EMAIL)) $errors[] = 'Invalid email format.';
    if (strlen($password) < 6) $errors[] = 'Password must be at least 6 characters.';
    if (!preg_match('/^[0-9]{10}$/', $phone)) $errors[] = 'Phone must be exactly 10 digits.';

    if (empty($errors)) {
        $conn = new mysqli('localhost', 'root', '', 'qbank');
        if ($conn->connect_error) die('DB connection failed: ' . $conn->connect_error);

        // Check duplicate email
        $stmt = $conn->prepare('SELECT id FROM users WHERE email = ?');
        $stmt->bind_param('s', $email);
        $stmt->execute();
        $stmt->store_result();
        if ($stmt->num_rows > 0) {
            $errors[] = 'Email is already registered.';
        } else {
            $hashed = password_hash($password, PASSWORD_DEFAULT);
            $ins = $conn->prepare(
                'INSERT INTO users (name, email, password, phone) VALUES (?, ?, ?, ?)'
            );
            $ins->bind_param('ssss', $name, $email, $hashed, $phone);
            if ($ins->execute()) {
                header('Location: welcome.php?name=' . urlencode($name));
                exit;
            } else {
                $errors[] = 'Registration failed: ' . $conn->error;
            }
            $ins->close();
        }
    }
}
```

```

        $stmt->close();
        $conn->close();
    }
}
?>
<!DOCTYPE html>
<html><head><meta charset="UTF-8"><title>Register</title>
<style>
    body{font-family:Arial;max-width:420px;margin:50px auto;padding:20px;background:#f7f7f7;}
    h2{color:#333;}
    input{width:100%;padding:10px;margin:6px 0 12px;box-sizing:border-box;
        border:1px solid #ccc;border-radius:4px;}
    button{padding:12px 20px;background:#4CAF50;color:#fff;border:none;border-radius:4px;cursor:pointer;}
    .err{background:#f8d7da;color:#721c24;padding:10px;border-radius:4px;margin-bottom:12px;}
</style></head><body>
<h2>User Registration</h2>
<?php if (!empty($errors)): ?>
    <div class="err"><?php foreach ($errors as $e) echo htmlspecialchars($e) . '<br>'; ?></div>
<?php endif; ?>
<form method="POST">
    <label>Name</label>
    <input type="text" name="name" value="<?php echo htmlspecialchars($name); ?>">
    <label>Email</label>
    <input type="email" name="email" value="<?php echo htmlspecialchars($email); ?>">
    <label>Password</label>
    <input type="password" name="password">
    <label>Phone (10 digits)</label>
    <input type="text" name="phone" value="<?php echo htmlspecialchars($phone); ?>">
    <button type="submit">Register</button>
</form>
</body></html>

<?php
// ===== FILE: welcome.php =====
$name = htmlspecialchars($_GET['name'] ?? 'User');
?>
<!DOCTYPE html>
<html><head><meta charset="UTF-8"><title>Welcome</title>
<style>
    body{font-family:Arial;text-align:center;margin-top:80px;}
    h1{color:#4CAF50;}
</style></head><body>
<h1>Welcome, <?php echo $name; ?>!</h1>
<p>Your account has been created successfully.</p>
<a href="register.php">Register another user</a>
</body></html>

```

Execution Instructions

Place this file inside C:\xampp\htdocs\<folder>\ (Windows) or /Applications/XAMPP/htdocs/<folder>/ (Mac). Start Apache and MySQL from the XAMPP Control Panel. Open phpMyAdmin (<http://localhost/phpmyadmin>) and run the SQL shown at the top of the file to create the database and table. Then visit <http://localhost/<folder>/<filename>> in your browser. You will create two files: register.php and welcome.php in the same folder.

Q11. Student Record Management System

XAMPP REQUIRED

Problem Statement

Develop a system where an admin can add, view, edit, and delete student details (Roll No, Name, Class, Marks). Validate inputs (Roll No unique, marks numeric). Display all students in an HTML table sorted by Roll No. Provide search by name or Roll No.

File(s)

students.php

Final Working Code

```
-- =====
-- SQL: run once in phpMyAdmin
-- =====
CREATE DATABASE IF NOT EXISTS qbank;
USE qbank;
CREATE TABLE IF NOT EXISTS students (
    roll_no INT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    class VARCHAR(50) NOT NULL,
    marks DECIMAL(5,2) NOT NULL
);

<?php
// ===== FILE: students.php =====
$conn = new mysqli('localhost', 'root', '', 'qbank');
if ($conn->connect_error) die('DB error: ' . $conn->connect_error);

$msg = '';
$edit_row = null;

// ----- DELETE -----
if (isset($_GET['delete'])) {
    $del = $conn->prepare('DELETE FROM students WHERE roll_no = ?');
    $del->bind_param('i', $_GET['delete']);
    $del->execute();
    header('Location: students.php');
    exit;
}

// ----- LOAD ROW FOR EDIT -----
if (isset($_GET['edit'])) {
    $stmt = $conn->prepare('SELECT * FROM students WHERE roll_no = ?');
    $stmt->bind_param('i', $_GET['edit']);
    $stmt->execute();
    $edit_row = $stmt->get_result()->fetch_assoc();
}

// ----- ADD or UPDATE -----
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $roll = (int)($_POST['roll_no'] ?? 0);
    $name = trim($_POST['name'] ?? '');
    $class = trim($_POST['class'] ?? '');
    $marks = $_POST['marks'] ?? '';
    $mode = $_POST['mode'] ?? 'add';

    if ($roll <= 0 || $name === '' || $class === '' || !is_numeric($marks)) {
        $msg = '<div class="err">All fields required; marks must be numeric; Roll No must be positive.</div>';
    } else {
        if ($mode === 'add') {
            $chk = $conn->prepare('SELECT roll_no FROM students WHERE roll_no = ?');
            $chk->bind_param('i', $roll);
            $chk->execute();
            $chk->store_result();
            if ($chk->num_rows > 0) {
                $msg = '<div class="err">Roll No already exists.</div>';
            } else {
                $ins = $conn->prepare('INSERT INTO students VALUES (?, ?, ?, ?)');
```



```

        $ins->bind_param('issd', $roll, $name, $class, $marks);
        $ins->execute();
        $msg = '<div class="ok">Student added.</div>';
    }
} else {
    $upd = $conn->prepare('UPDATE students SET name=?, class=?, marks=? WHERE roll_no=?');
    $upd->bind_param('ssdi', $name, $class, $marks, $roll);
    $upd->execute();
    $msg = '<div class="ok">Student updated.</div>';
}
}
}

// ----- SEARCH -----
$search = trim($_GET['q'] ?? '');
if ($search !== '') {
    $like = '%' . $search . '%';
    $stmt = $conn->prepare('SELECT * FROM students WHERE name LIKE ? OR roll_no = ? ORDER BY roll_no');
    $rollSearch = is_numeric($search) ? (int)$search : 0;
    $stmt->bind_param('si', $like, $rollSearch);
    $stmt->execute();
    $rows = $stmt->get_result();
} else {
    $rows = $conn->query('SELECT * FROM students ORDER BY roll_no');
}
?>
<!DOCTYPE html>
<html><head><meta charset="UTF-8"><title>Student Records</title>
<style>
    body{font-family:Arial;max-width:900px;margin:30px auto;padding:0 20px;}
    h2{color:#333;}
    form.inline{display:flex;gap:8px;flex-wrap:wrap;align-items:center;
        background:#f4f4f4;padding:15px;border-radius:6px;margin-bottom:15px;}
    input{padding:8px;border:1px solid #ccc;border-radius:4px;}
    button{padding:8px 14px;background:#4CAF50;color:#fff;border:none;border-radius:4px;cursor:pointer;}
    table{width:100%;border-collapse:collapse;margin-top:10px;}
    th,td{border:1px solid #ddd;padding:10px;text-align:left;}
    th{background:#2196F3;color:#fff;}
    tr:nth-child(even){background:#f9f9f9;}
    .err{background:#f8d7da;color:#721c24;padding:10px;border-radius:4px;margin:10px 0;}
    .ok{background:#d4edda;color:#155724;padding:10px;border-radius:4px;margin:10px 0;}
    a.act{margin:0 5px;text-decoration:none;}
</style></head><body>
<h2>Student Record Management</h2>
<?php echo $msg; ?>

<h3><?php echo $edit_row ? 'Edit Student' : 'Add Student'; ?></h3>
<form method="POST" class="inline">
    <input type="hidden" name="mode" value="<?php echo $edit_row ? 'edit' : 'add'; ?>">
    <input type="number" name="roll_no" placeholder="Roll No" required
        value="<?php echo $edit_row['roll_no'] ?? ''; ?>"
        <?php echo $edit_row ? 'readonly' : ''; ?>
    <input type="text" name="name" placeholder="Name" required
        value="<?php echo htmlspecialchars($edit_row['name'] ?? ''); ?>">
    <input type="text" name="class" placeholder="Class" required
        value="<?php echo htmlspecialchars($edit_row['class'] ?? ''); ?>">
    <input type="number" step="0.01" name="marks" placeholder="Marks" required
        value="<?php echo $edit_row['marks'] ?? ''; ?>">
    <button type="submit"><?php echo $edit_row ? 'Update' : 'Add'; ?></button>
    <?php if ($edit_row) echo '<a href="students.php">Cancel</a>'; ?>
</form>

<form method="GET" class="inline">
    <input type="text" name="q" placeholder="Search by name or roll no"
        value="<?php echo htmlspecialchars($search); ?>">
    <button type="submit">Search</button>
    <?php if ($search) echo '<a href="students.php">Clear</a>'; ?>
</form>

<table>
    <tr><th>Roll No</th><th>Name</th><th>Class</th><th>Marks</th><th>Actions</th></tr>
    <?php while ($r = $rows->fetch_assoc()): ?>
    <tr>

```

```

<td><?php echo $r['roll_no']; ?></td>
<td><?php echo htmlspecialchars($r['name']); ?></td>
<td><?php echo htmlspecialchars($r['class']); ?></td>
<td><?php echo $r['marks']; ?></td>
<td>
    <a class="act" href="?edit=<?php echo $r['roll_no']; ?>">Edit</a> |
    <a class="act" href="?delete=<?php echo $r['roll_no']; ?>"
        onclick="return confirm('Delete this student?');">Delete</a>
</td>
</tr>
<?php endwhile; ?>
</table>
</body></html>
<?php $conn->close(); ?>

```

Execution Instructions

Place this file inside C:\xampp\htdocs\<folder>\ (Windows) or /Applications/XAMPP/htdocs/<folder>/ (Mac). Start Apache and MySQL from the XAMPP Control Panel. Open phpMyAdmin (<http://localhost/phpmyadmin>) and run the SQL shown at the top of the file to create the database and table. Then visit <http://localhost/<folder>/<filename>> in your browser. Single file handles add, view, edit, delete, and search.

Q12. Product Inventory Management

XAMPP REQUIRED

Problem Statement

Create an inventory system for a small shop. The owner can add new products (auto-generated unique Product ID, Name, Price, Quantity). Validate that Price and Quantity are positive numbers. Display all products in a table with options to update stock or delete entries.

File(s)

inventory.php

Final Working Code

```
-- =====
-- SQL
-- =====
CREATE DATABASE IF NOT EXISTS qbank;
USE qbank;
CREATE TABLE IF NOT EXISTS products (
    product_id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    price DECIMAL(10,2) NOT NULL,
    quantity INT NOT NULL
);

<?php
// ===== FILE: inventory.php =====
$conn = new mysqli('localhost', 'root', '', 'qbank');
if ($conn->connect_error) die('DB error: ' . $conn->connect_error);

$msg = '';

// DELETE
if (isset($_GET['delete'])) {
    $stmt = $conn->prepare('DELETE FROM products WHERE product_id = ?');
    $stmt->bind_param('i', $_GET['delete']);
    $stmt->execute();
    header('Location: inventory.php');
    exit;
}

// ADD
if (isset($_POST['add'])) {
    $name = trim($_POST['name'] ?? '');
    $price = $_POST['price'] ?? '';
    $qty = $_POST['quantity'] ?? '';
    if ($name === '' || !is_numeric($price) || $price <= 0 ||
        !is_numeric($qty) || $qty < 0) {
        $msg = '<div class="err">Name required; price must be > 0; quantity must be a non-negative number.</div>';
    } else {
        $stmt = $conn->prepare('INSERT INTO products (name, price, quantity) VALUES (?, ?, ?)');
        $stmt->bind_param('sdi', $name, $price, $qty);
        $stmt->execute();
        $msg = '<div class="ok">Product added (ID = ' . $conn->insert_id . ').</div>';
    }
}

// UPDATE STOCK
if (isset($_POST['update_stock'])) {
    $id = (int)$_POST['product_id'];
    $qty = $_POST['new_qty'];
    if (!is_numeric($qty) || $qty < 0) {
        $msg = '<div class="err">Quantity must be a non-negative number.</div>';
    } else {
        $stmt = $conn->prepare('UPDATE products SET quantity = ? WHERE product_id = ?');
        $stmt->bind_param('ii', $qty, $id);
        $stmt->execute();
        $msg = '<div class="ok">Stock updated for product ID ' . $id . '</div>';
    }
}
```

```

$rows = $conn->query('SELECT * FROM products ORDER BY product_id');
?>
<!DOCTYPE html>
<html><head><meta charset="UTF-8"><title>Inventory</title>
<style>
    body{font-family:Arial;max-width:900px;margin:30px auto;padding:0 20px;}
    h2,h3{color:#333;}
    form.inline{display:flex;gap:8px;flex-wrap:wrap;align-items:center;
        background:#f4f4f4;padding:15px;border-radius:6px;margin-bottom:15px;}
    input{padding:8px;border:1px solid #ccc;border-radius:4px;}
    button{padding:8px 14px;background:#4CAF50;color:#fff;border:none;border-radius:4px;cursor:pointer;}
    table{width:100%;border-collapse:collapse;}
    th,td{border:1px solid #ddd;padding:10px;text-align:left;}
    th{background:#2196F3;color:#fff;}
    tr:nth-child(even){background:#f9f9f9;}
    .err{background:#f8d7da;color:#721c24;padding:10px;border-radius:4px;margin:10px 0;}
    .ok{background:#d4edda;color:#155724;padding:10px;border-radius:4px;margin:10px 0;}
</style></head><body>
<h2>Product Inventory Management</h2>
<?php echo $msg; ?>

<h3>Add Product</h3>
<form method="POST" class="inline">
    <input type="text" name="name" placeholder="Product Name" required>
    <input type="number" step="0.01" name="price" placeholder="Price" required>
    <input type="number" name="quantity" placeholder="Quantity" required>
    <button type="submit" name="add">Add Product</button>
</form>

<h3>Current Stock</h3>
<table>
    <tr><th>ID</th><th>Name</th><th>Price</th><th>Quantity</th><th>Update Stock</th><th>Action</th></tr>
    <?php while ($r = $rows->fetch_assoc()): ?>
    <tr>
        <td><?php echo $r['product_id']; ?></td>
        <td><?php echo htmlspecialchars($r['name']); ?></td>
        <td>$<?php echo number_format($r['price'], 2); ?></td>
        <td><?php echo $r['quantity']; ?></td>
        <td>
            <form method="POST" class="inline" style="margin:0;padding:0;background:none;">
                <input type="hidden" name="product_id" value="<?php echo $r['product_id']; ?>">
                <input type="number" name="new_qty" value="<?php echo $r['quantity']; ?>"
                    style="width:80px;">
                <button type="submit" name="update_stock">Update</button>
            </form>
        </td>
        <td>
            <a href="?delete=<?php echo $r['product_id']; ?>"
                onclick="return confirm('Delete this product?');">Delete</a>
        </td>
    </tr>
    <?php endwhile; ?>
</table>
</body></html>
<?php $conn->close(); ?>

```

Execution Instructions

Place this file inside C:\xampp\htdocs\<folder> (Windows) or /Applications/XAMPP/htdocs/<folder> (Mac). Start Apache and MySQL from the XAMPP Control Panel. Open phpMyAdmin (<http://localhost/phpmyadmin>) and run the SQL shown at the top of the file to create the database and table. Then visit <http://localhost/<folder>/<filename>> in your browser.

Q13. Task Manager Application

XAMPP REQUIRED

Problem Statement

Build a task manager where users can add tasks (Task Name, Description, Due Date), validating that Due Date is in the future. Mark tasks as complete (update status). Delete tasks permanently. Display pending and completed tasks in separate lists.

File(s)

tasks.php

Final Working Code

```
-- =====
-- SQL
-- =====
CREATE DATABASE IF NOT EXISTS qbank;
USE qbank;
CREATE TABLE IF NOT EXISTS tasks (
    id INT AUTO_INCREMENT PRIMARY KEY,
    task_name VARCHAR(150) NOT NULL,
    description TEXT,
    due_date DATE NOT NULL,
    status ENUM('pending','completed') DEFAULT 'pending',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

<?php
// ===== FILE: tasks.php =====
$conn = new mysqli('localhost', 'root', '', 'qbank');
if ($conn->connect_error) die('DB error: ' . $conn->connect_error);

$msg = '';

if (isset($_GET['delete'])) {
    $stmt = $conn->prepare('DELETE FROM tasks WHERE id = ?');
    $stmt->bind_param('i', $_GET['delete']);
    $stmt->execute();
    header('Location: tasks.php');
    exit;
}

if (isset($_GET['complete'])) {
    $stmt = $conn->prepare('UPDATE tasks SET status='completed' WHERE id = ?');
    $stmt->bind_param('i', $_GET['complete']);
    $stmt->execute();
    header('Location: tasks.php');
    exit;
}

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $name = trim($_POST['task_name'] ?? '');
    $desc = trim($_POST['description'] ?? '');
    $due = $_POST['due_date'] ?? '';

    if ($name === '' || $due === '') {
        $msg = '<div class="err">Task name and due date are required.</div>';
    } elseif (strtotime($due) < strtotime(date('Y-m-d'))) {
        $msg = '<div class="err">Due date must be today or in the future.</div>';
    } else {
        $stmt = $conn->prepare(
            'INSERT INTO tasks (task_name, description, due_date) VALUES (?, ?, ?)'
        );
        $stmt->bind_param('sss', $name, $desc, $due);
        $stmt->execute();
        $msg = '<div class="ok">Task added.</div>';
    }
}

$pending = $conn->query("SELECT * FROM tasks WHERE status='pending' ORDER BY due_date");
```

```

$completed = $conn->query("SELECT * FROM tasks WHERE status='completed' ORDER BY due_date DESC");
?>
<!DOCTYPE html>
<html><head><meta charset="UTF-8"><title>Task Manager</title>
<style>
body{font-family:Arial;max-width:900px;margin:30px auto;padding:0 20px;}
h2,h3{color:#333;}
form{background:#f4f4f4;padding:15px;border-radius:6px;margin-bottom:20px;
display:flex;gap:8px;flex-wrap:wrap;align-items:center;}
input,textarea{padding:8px;border:1px solid #ccc;border-radius:4px;}
textarea{width:100%;}
button{padding:8px 14px;background:#4CAF50;color:#fff;border:none;border-radius:4px;cursor:pointer;}
.lists{display:flex;gap:20px;}
.lists>div{flex:1;background:#fafafa;padding:15px;border-radius:6px;}
.task{background:#fff;padding:10px;margin:6px 0;border-left:4px solid #2196F3;border-radius:4px;}
.task.done{border-left-color:#4CAF50;text-decoration:line-through;color:#666;}
.err{background:#f8d7da;color:#721c24;padding:10px;border-radius:4px;margin:10px 0;}
.ok{background:#d4edda;color:#155724;padding:10px;border-radius:4px;margin:10px 0;}
</style></head><body>
<h2>Task Manager</h2>
<?php echo $msg; ?>

<form method="POST">
<input type="text" name="task_name" placeholder="Task name" required>
<input type="date" name="due_date" required min="<?php echo date('Y-m-d'); ?>">
<textarea name="description" placeholder="Description (optional)" rows="2"></textarea>
<button type="submit">Add Task</button>
</form>

<div class="lists">
<div>
<h3>Pending Tasks</h3>
<?php while ($t = $pending->fetch_assoc()): ?>
<div class="task">
<strong><?php echo htmlspecialchars($t['task_name']); ?></strong>
(due <?php echo $t['due_date']; ?>)<br>
<small><?php echo nl2br(htmlspecialchars($t['description'])); ?></small><br>
<a href="?complete=<?php echo $t['id']; ?>">Mark complete</a> |
<a href="?delete=<?php echo $t['id']; ?>"
onclick="return confirm('Delete?');">Delete</a>
</div>
<?php endwhile; ?>
</div>
<div>
<h3>Completed Tasks</h3>
<?php while ($t = $completed->fetch_assoc()): ?>
<div class="task done">
<strong><?php echo htmlspecialchars($t['task_name']); ?></strong>
(was due <?php echo $t['due_date']; ?>)
<a href="?delete=<?php echo $t['id']; ?>"
onclick="return confirm('Delete?');" style="float:right;">Delete</a>
</div>
<?php endwhile; ?>
</div>
</div>
</body></html>
<?php $conn->close(); ?>

```

Execution Instructions

Place this file inside C:\xampp\htdocs\<folder> (Windows) or /Applications/XAMPP/htdocs/<folder> (Mac). Start Apache and MySQL from the XAMPP Control Panel. Open phpMyAdmin (<http://localhost/phpmyadmin>) and run the SQL shown at the top of the file to create the database and table. Then visit <http://localhost/<folder>/<filename>> in your browser.

Q14. Library Book Tracking System

XAMPP REQUIRED

Problem Statement

Design a library system where the librarian can add books (unique Book ID, Title, Author, Category) with input validation. Users can search books by title or author. Provide options to update book details or remove books from the database.

File(s)

library.php

Final Working Code

```
-- =====
-- SQL
-- =====
CREATE DATABASE IF NOT EXISTS qbank;
USE qbank;
CREATE TABLE IF NOT EXISTS books (
    book_id VARCHAR(20) PRIMARY KEY,
    title VARCHAR(200) NOT NULL,
    author VARCHAR(150) NOT NULL,
    category VARCHAR(50) NOT NULL
);

<?php
// ===== FILE: library.php =====
$conn = new mysqli('localhost', 'root', '', 'qbank');
if ($conn->connect_error) die('DB error: ' . $conn->connect_error);

$msg = '';
$edit_row = null;

if (isset($_GET['delete'])) {
    $stmt = $conn->prepare('DELETE FROM books WHERE book_id = ?');
    $stmt->bind_param('s', $_GET['delete']);
    $stmt->execute();
    header('Location: library.php');
    exit;
}

if (isset($_GET['edit'])) {
    $stmt = $conn->prepare('SELECT * FROM books WHERE book_id = ?');
    $stmt->bind_param('s', $_GET['edit']);
    $stmt->execute();
    $edit_row = $stmt->get_result()->fetch_assoc();
}

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $id = trim($_POST['book_id']);
    $title = trim($_POST['title']);
    $auth = trim($_POST['author']);
    $cat = trim($_POST['category']);
    $mode = $_POST['mode'];

    if ($id === '' || $title === '' || $auth === '' || $cat === '') {
        $msg = '<div class="err">All fields are required.</div>';
    } elseif ($mode === 'add') {
        $chk = $conn->prepare('SELECT book_id FROM books WHERE book_id = ?');
        $chk->bind_param('s', $id);
        $chk->execute();
        $chk->store_result();
        if ($chk->num_rows > 0) {
            $msg = '<div class="err">Book ID already exists.</div>';
        } else {
            $ins = $conn->prepare('INSERT INTO books VALUES (?, ?, ?, ?)');
            $ins->bind_param('ssss', $id, $title, $auth, $cat);
            $ins->execute();
            $msg = '<div class="ok">Book added.</div>';
        }
    }
}
```

```

    } else { // edit
        $upd = $conn->prepare(
            'UPDATE books SET title=?, author=?, category=? WHERE book_id=?'
        );
        $upd->bind_param('ssss', $title, $auth, $cat, $id);
        $upd->execute();
        $msg = '<div class="ok">Book updated.</div>';
    }
}

$search = trim($_GET['q'] ?? '');
if ($search !== '') {
    $like = '%' . $search . '%';
    $stmt = $conn->prepare(
        'SELECT * FROM books WHERE title LIKE ? OR author LIKE ? ORDER BY title'
    );
    $stmt->bind_param('ss', $like, $like);
    $stmt->execute();
    $rows = $stmt->get_result();
} else {
    $rows = $conn->query('SELECT * FROM books ORDER BY title');
}
?>
<!DOCTYPE html>
<html><head><meta charset="UTF-8"><title>Library</title>
<style>
    body{font-family:Arial;max-width:950px;margin:30px auto;padding:0 20px;}
    h2,h3{color:#333;}
    form.inline{display:flex;gap:8px;flex-wrap:wrap;align-items:center;
        background:#f4f4f4;padding:15px;border-radius:6px;margin-bottom:15px;}
    input{padding:8px;border:1px solid #ccc;border-radius:4px;}
    button{padding:8px 14px;background:#4CAF50;color:#fff;border:none;border-radius:4px;cursor:pointer;}
    table{width:100%;border-collapse:collapse;}
    th,td{border:1px solid #ddd;padding:10px;text-align:left;}
    th{background:#2196F3;color:#fff;}
    tr:nth-child(even){background:#f9f9f9;}
    .err{background:#f8d7da;color:#721c24;padding:10px;border-radius:4px;margin:10px 0;}
    .ok{background:#d4edda;color:#155724;padding:10px;border-radius:4px;margin:10px 0;}
</style></head><body>
<h2>Library Book Tracking</h2>
<?php echo $msg; ?>

<h3><?php echo $edit_row ? 'Edit Book' : 'Add Book'; ?></h3>
<form method="POST" class="inline">
    <input type="hidden" name="mode" value="<?php echo $edit_row ? 'edit' : 'add'; ?>">
    <input type="text" name="book_id" placeholder="Book ID" required
        value="<?php echo htmlspecialchars($edit_row['book_id'] ?? ''); ?>"
        <?php echo $edit_row ? 'readonly' : ''; ?>
    <input type="text" name="title" placeholder="Title" required
        value="<?php echo htmlspecialchars($edit_row['title'] ?? ''); ?>">
    <input type="text" name="author" placeholder="Author" required
        value="<?php echo htmlspecialchars($edit_row['author'] ?? ''); ?>">
    <input type="text" name="category" placeholder="Category" required
        value="<?php echo htmlspecialchars($edit_row['category'] ?? ''); ?>">
    <button type="submit"><?php echo $edit_row ? 'Update' : 'Add'; ?></button>
    <?php if ($edit_row) echo '<a href="library.php">Cancel</a>'; ?>
</form>

<form method="GET" class="inline">
    <input type="text" name="q" placeholder="Search by title or author"
        value="<?php echo htmlspecialchars($search); ?>">
    <button type="submit">Search</button>
    <?php if ($search) echo '<a href="library.php">Clear</a>'; ?>
</form>

<table>
<tr><th>ID</th><th>Title</th><th>Author</th><th>Category</th><th>Actions</th></tr>
<?php while ($r = $rows->fetch_assoc()): ?>
<tr>
    <td><?php echo htmlspecialchars($r['book_id']); ?></td>
    <td><?php echo htmlspecialchars($r['title']); ?></td>
    <td><?php echo htmlspecialchars($r['author']); ?></td>
    <td><?php echo htmlspecialchars($r['category']); ?></td>

```



```

        <td>
            <a href="?edit=?php echo urlencode($r['book_id']); ?>">Edit</a> |
            <a href="?delete=?php echo urlencode($r['book_id']); ?>"
                onclick="return confirm('Delete?');">Delete</a>
        </td>
    </tr>
</table>
</body></html>
<?php $conn->close(); ?>

```

Execution Instructions

Place this file inside C:\xampp\htdocs\<folder>\ (Windows) or /Applications/XAMPP/htdocs/<folder>/ (Mac). Start Apache and MySQL from the XAMPP Control Panel. Open phpMyAdmin (<http://localhost/phpmyadmin>) and run the SQL shown at the top of the file to create the database and table. Then visit <http://localhost/<folder>/<filename>> in your browser.

Q15. Employee Attendance System

XAMPP REQUIRED

Problem Statement

Create a system where employees mark attendance (Present/Absent) for the current date. Validate that an employee cannot mark attendance twice for the same day. Admin can view attendance reports filtered by date or employee. Store records in MySQL with employee_id, date, and status.

File(s)

attendance.php

Final Working Code

```
-- =====
-- SQL
-- =====
CREATE DATABASE IF NOT EXISTS qbank;
USE qbank;
CREATE TABLE IF NOT EXISTS employees (
    employee_id INT PRIMARY KEY,
    name VARCHAR(100) NOT NULL
);
CREATE TABLE IF NOT EXISTS attendance (
    id INT AUTO_INCREMENT PRIMARY KEY,
    employee_id INT NOT NULL,
    date DATE NOT NULL,
    status ENUM('Present', 'Absent') NOT NULL,
    UNIQUE KEY unique_emp_date (employee_id, date),
    FOREIGN KEY (employee_id) REFERENCES employees(employee_id)
);

-- Sample employees so the system has data to work with
INSERT IGNORE INTO employees VALUES
    (101, 'Alice'),
    (102, 'Bob'),
    (103, 'Charlie'),
    (104, 'Diana');

<?php
// ===== FILE: attendance.php =====
$conn = new mysqli('localhost', 'root', '', 'qbank');
if ($conn->connect_error) die('DB error: ' . $conn->connect_error);

$msg = '';
$today = date('Y-m-d');

if ($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST['mark'])) {
    $eid = (int)$_POST['employee_id'];
    $status = $_POST['status'];
    if (!in_array($status, ['Present', 'Absent'])) {
        $msg = '<div class="err">Invalid status.</div>';
    } else {
        // Check duplicate
        $chk = $conn->prepare('SELECT id FROM attendance WHERE employee_id=? AND date=?');
        $chk->bind_param('is', $eid, $today);
        $chk->execute();
        $chk->store_result();
        if ($chk->num_rows > 0) {
            $msg = '<div class="err">Attendance already marked for this employee today.</div>';
        } else {
            $ins = $conn->prepare(
                'INSERT INTO attendance (employee_id, date, status) VALUES (?, ?, ?)'
            );
            $ins->bind_param('iss', $eid, $today, $status);
            $ins->execute();
            $msg = '<div class="ok">Attendance recorded.</div>';
        }
    }
}
}
```

```

// Build report query with optional filters
$f_emp = $_GET['f_emp'] ?? '';
$f_date = $_GET['f_date'] ?? '';

$sql = 'SELECT a.id, a.employee_id, e.name, a.date, a.status
        FROM attendance a JOIN employees e ON a.employee_id = e.employee_id WHERE 1=1';
$types = '';
$args = [];
if ($f_emp !== '') {
    $sql .= ' AND a.employee_id = ?';
    $types .= 'i';
    $args[] = (int)$f_emp;
}
if ($f_date !== '') {
    $sql .= ' AND a.date = ?';
    $types .= 's';
    $args[] = $f_date;
}
$sql .= ' ORDER BY a.date DESC, a.employee_id';

$stmt = $conn->prepare($sql);
if ($types !== '') $stmt->bind_param($types, ...$args);
$stmt->execute();
$rows = $stmt->get_result();

$emps = $conn->query('SELECT * FROM employees ORDER BY employee_id');
?>
<!DOCTYPE html>
<html><head><meta charset="UTF-8"><title>Attendance</title>
<style>
    body{font-family:Arial;max-width:900px;margin:30px auto;padding:0 20px;}
    h2,h3{color:#333;}
    form.inline{display:flex;gap:8px;flex-wrap:wrap;align-items:center;
        background:#f4f4f4;padding:15px;border-radius:6px;margin-bottom:15px;}
    input,select{padding:8px;border:1px solid #ccc;border-radius:4px;}
    button{padding:8px 14px;background:#4CAF50;color:#fff;border:none;border-radius:4px;cursor:pointer;}
    table{width:100%;border-collapse:collapse;}
    th,td{border:1px solid #ddd;padding:10px;text-align:left;}
    th{background:#2196F3;color:#fff;}
    .err{background:#f8d7da;color:#721c24;padding:10px;border-radius:4px;margin:10px 0;}
    .ok{background:#d4edda;color:#155724;padding:10px;border-radius:4px;margin:10px 0;}
</style></head><body>
<h2>Employee Attendance (today is <?php echo $today; ?>)</h2>
<?php echo $msg; ?>

<h3>Mark Attendance</h3>
<form method="POST" class="inline">
    <select name="employee_id" required>
        <option value="">-- Select Employee --</option>
        <?php while ($e = $emps->fetch_assoc()): ?>
            <option value="<?php echo $e['employee_id']; ?>">
                <?php echo $e['employee_id'] . ' - ' . htmlspecialchars($e['name']); ?>
            </option>
        <?php endwhile; ?>
    </select>
    <select name="status">
        <option>Present</option><option>Absent</option>
    </select>
    <button type="submit" name="mark">Mark</button>
</form>

<h3>Attendance Report</h3>
<form method="GET" class="inline">
    <input type="text" name="f_emp" placeholder="Employee ID"
        value="<?php echo htmlspecialchars($f_emp); ?>">
    <input type="date" name="f_date" value="<?php echo htmlspecialchars($f_date); ?>">
    <button type="submit">Filter</button>
    <?php if ($f_emp || $f_date) echo ' <a href="attendance.php">Clear</a>'; ?>
</form>

<table>
    <tr><th>ID</th><th>Employee ID</th><th>Name</th><th>Date</th><th>Status</th></tr>
    <?php while ($r = $rows->fetch_assoc()): ?>

```

```

<tr>
  <td><?php echo $r['id']; ?></td>
  <td><?php echo $r['employee_id']; ?></td>
  <td><?php echo htmlspecialchars($r['name']); ?></td>
  <td><?php echo $r['date']; ?></td>
  <td><?php echo $r['status']; ?></td>
</tr>
<?php endwhile; ?>
</table>
</body></html>
<?php $conn->close(); ?>

```

Execution Instructions

Place this file inside C:\xampp\htdocs\<folder>\ (Windows) or /Applications/XAMPP/htdocs/<folder>/ (Mac). Start Apache and MySQL from the XAMPP Control Panel. Open phpMyAdmin (<http://localhost/phpmyadmin>) and run the SQL shown at the top of the file to create the database and table. Then visit <http://localhost/<folder>/<filename>> in your browser.

Q16. Online Feedback Form

XAMPP REQUIRED

Problem Statement

Develop a feedback form for a college website. Students submit feedback (Name, Email, Feedback Message). Validate inputs (valid email, message at least 20 characters). Store feedback in MySQL. Display all feedback in a paginated table (5 entries per page).

File(s)

feedback.php

Final Working Code

```
-- =====
-- SQL
-- =====
CREATE DATABASE IF NOT EXISTS qbank;
USE qbank;
CREATE TABLE IF NOT EXISTS feedback (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(150) NOT NULL,
    message TEXT NOT NULL,
    submitted_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

<?php
// ===== FILE: feedback.php =====
$conn = new mysqli('localhost', 'root', '', 'qbank');
if ($conn->connect_error) die('DB error: ' . $conn->connect_error);

$msg = '';

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $name = trim($_POST['name'] ?? '');
    $em   = trim($_POST['email'] ?? '');
    $txt  = trim($_POST['message'] ?? '');

    if ($name === '' || !filter_var($em, FILTER_VALIDATE_EMAIL) || strlen($txt) < 20) {
        $msg = '<div class="err">Name required, valid email required, message must be at least 20 characters.</div>';
    } else {
        $stmt = $conn->prepare('INSERT INTO feedback (name, email, message) VALUES (?, ?, ?)');
        $stmt->bind_param('sss', $name, $em, $txt);
        $stmt->execute();
        $msg = '<div class="ok">Thanks! Your feedback has been recorded.</div>';
    }
}

// PAGINATION
$per_page = 5;
$page = max(1, (int)($_GET['page'] ?? 1));
$offset = ($page - 1) * $per_page;

$total = $conn->query('SELECT COUNT(*) AS c FROM feedback')->fetch_assoc()['c'];
$pages = max(1, (int)ceil($total / $per_page));

$rows = $conn->query("SELECT * FROM feedback ORDER BY submitted_at DESC LIMIT $per_page OFFSET $offset");
?>
<!DOCTYPE html>
<html><head><meta charset="UTF-8"><title>Feedback</title>
<style>
    body{font-family:Arial;max-width:800px;margin:30px auto;padding:0 20px;}
    h2,h3{color:#333;}
    form{background:#f4f4f4;padding:15px;border-radius:6px;margin-bottom:20px;}
    input,textarea{width:100%;padding:8px;margin:5px 0;border:1px solid #ccc;
        border-radius:4px;box-sizing:border-box;}
    button{padding:10px 20px;background:#4CAF50;color:#fff;border:none;
        border-radius:4px;cursor:pointer;}
    table{width:100%;border-collapse:collapse;}
    th,td{border:1px solid #ddd;padding:10px;text-align:left;vertical-align:top;}
```

```

th{background:#2196F3;color:#fff;}
tr:nth-child(even){background:#f9f9f9;}
.err{background:#f8d7da;color:#721c24;padding:10px;border-radius:4px;margin:10px 0;}
.ok{background:#d4edda;color:#155724;padding:10px;border-radius:4px;margin:10px 0;}
.pager{margin-top:15px;text-align:center;}
.pager a,.pager span{display:inline-block;padding:6px 12px;margin:0 3px;
border:1px solid #ddd;border-radius:4px;text-decoration:none;color:#333;}
.pager .current{background:#2196F3;color:#fff;border-color:#2196F3;}
</style></head><body>
<h2>Submit Feedback</h2>
<?php echo $msg; ?>
<form method="POST">
  <input type="text" name="name" placeholder="Your Name" required>
  <input type="email" name="email" placeholder="Your Email" required>
  <textarea name="message" rows="4" placeholder="Your feedback (min 20 chars)" required></textarea>
  <button type="submit">Submit Feedback</button>
</form>

<h3>Submitted Feedback (<?php echo $total; ?>)</h3>
<table>
  <tr><th>Name</th><th>Email</th><th>Message</th><th>When</th></tr>
  <?php while ($r = $rows->fetch_assoc()): ?>
    <tr>
      <td><?php echo htmlspecialchars($r['name']); ?></td>
      <td><?php echo htmlspecialchars($r['email']); ?></td>
      <td><?php echo nl2br(htmlspecialchars($r['message'])); ?></td>
      <td><?php echo $r['submitted_at']; ?></td>
    </tr>
  <?php endwhile; ?>
</table>

<div class="pager">
  <?php for ($i = 1; $i <= $pages; $i++): ?>
    <?php if ($i === $page): ?>
      <span class="current"><?php echo $i; ?></span>
    <?php else: ?>
      <a href="?page=<?php echo $i; ?>"><?php echo $i; ?></a>
    <?php endif; ?>
  <?php endfor; ?>
</div>
</body></html>
<?php $conn->close(); ?>

```

Execution Instructions

Place this file inside C:\xampp\htdocs\<folder>\ (Windows) or /Applications/XAMPP/htdocs/<folder>/ (Mac). Start Apache and MySQL from the XAMPP Control Panel. Open phpMyAdmin (<http://localhost/phpmyadmin>) and run the SQL shown at the top of the file to create the database and table. Then visit <http://localhost/<folder>/<filename>> in your browser.

Q17. Movie Database Manager

XAMPP REQUIRED

Problem Statement

Build a system where the admin can add movies (Title, Director, Release Year, Genre). Validate that Release Year is between 1900 and 2025. Users can filter movies by genre using a dropdown. Admin can edit or delete movie entries.

File(s)

movies.php

Final Working Code

```
-- =====
-- SQL
-- =====
CREATE DATABASE IF NOT EXISTS qbank;
USE qbank;
CREATE TABLE IF NOT EXISTS movies (
    id INT AUTO_INCREMENT PRIMARY KEY,
    title VARCHAR(200) NOT NULL,
    director VARCHAR(150) NOT NULL,
    release_year SMALLINT NOT NULL,
    genre VARCHAR(50) NOT NULL
);

<?php
// ===== FILE: movies.php =====
$conn = new mysqli('localhost', 'root', '', 'qbank');
if ($conn->connect_error) die('DB error: ' . $conn->connect_error);

$msg = '';
$edit_row = null;
$genres_list = ['Action', 'Comedy', 'Drama', 'Horror', 'Sci-Fi', 'Romance', 'Thriller', 'Documentary'];

if (isset($_GET['delete'])) {
    $stmt = $conn->prepare('DELETE FROM movies WHERE id = ?');
    $stmt->bind_param('i', $_GET['delete']);
    $stmt->execute();
    header('Location: movies.php');
    exit;
}

if (isset($_GET['edit'])) {
    $stmt = $conn->prepare('SELECT * FROM movies WHERE id = ?');
    $stmt->bind_param('i', $_GET['edit']);
    $stmt->execute();
    $edit_row = $stmt->get_result()->fetch_assoc();
}

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $title = trim($_POST['title']);
    $dir = trim($_POST['director']);
    $year = (int)$_POST['release_year'];
    $genre = $_POST['genre'];
    $mode = $_POST['mode'];

    if ($title === '' || $dir === '' || !in_array($genre, $genres_list)) {
        $msg = '<div class="err">All fields required and genre must be valid.</div>';
    } elseif ($year < 1900 || $year > 2025) {
        $msg = '<div class="err">Release year must be between 1900 and 2025.</div>';
    } elseif ($mode === 'add') {
        $ins = $conn->prepare(
            'INSERT INTO movies (title, director, release_year, genre) VALUES (?, ?, ?, ?)'
        );
        $ins->bind_param('ssis', $title, $dir, $year, $genre);
        $ins->execute();
        $msg = '<div class="ok">Movie added.</div>';
    } else {
        $id = (int)$_POST['id'];
```

```

        $upd = $conn->prepare(
            'UPDATE movies SET title=?, director=?, release_year=?, genre=? WHERE id=?'
        );
        $upd->bind_param('ssisi', $title, $dir, $year, $genre, $id);
        $upd->execute();
        $msg = '<div class="ok">Movie updated.</div>';
    }
}

$filter = $_GET['filter'] ?? '';
if ($filter !== '' && in_array($filter, $genres_list)) {
    $stmt = $conn->prepare('SELECT * FROM movies WHERE genre = ? ORDER BY release_year DESC');
    $stmt->bind_param('s', $filter);
    $stmt->execute();
    $rows = $stmt->get_result();
} else {
    $rows = $conn->query('SELECT * FROM movies ORDER BY release_year DESC');
}
?>
<!DOCTYPE html>
<html><head><meta charset="UTF-8"><title>Movies</title>
<style>
    body{font-family:Arial;max-width:950px;margin:30px auto;padding:0 20px;}
    h2,h3{color:#333;}
    form.inline{display:flex;gap:8px;flex-wrap:wrap;align-items:center;
        background:#f4f4f4;padding:15px;border-radius:6px;margin-bottom:15px;}
    input,select{padding:8px;border:1px solid #ccc;border-radius:4px;}
    button{padding:8px 14px;background:#4CAF50;color:#fff;border:none;border-radius:4px;cursor:pointer;}
    table{width:100%;border-collapse:collapse;}
    th,td{border:1px solid #ddd;padding:10px;text-align:left;}
    th{background:#2196F3;color:#fff;}
    tr:nth-child(even){background:#f9f9f9;}
    .err{background:#f8d7da;color:#721c24;padding:10px;border-radius:4px;margin:10px 0;}
    .ok{background:#d4edda;color:#155724;padding:10px;border-radius:4px;margin:10px 0;}
</style></head><body>
<h2>Movie Database Manager</h2>
<?php echo $msg; ?>

<h3><?php echo $edit_row ? 'Edit Movie' : 'Add Movie'; ?></h3>
<form method="POST" class="inline">
    <input type="hidden" name="mode" value="<?php echo $edit_row ? 'edit' : 'add'; ?>">
    <?php if ($edit_row): ?>
        <input type="hidden" name="id" value="<?php echo $edit_row['id']; ?>">
    <?php endif; ?>
    <input type="text" name="title" placeholder="Title" required
        value="<?php echo htmlspecialchars($edit_row['title'] ?? ''); ?>">
    <input type="text" name="director" placeholder="Director" required
        value="<?php echo htmlspecialchars($edit_row['director'] ?? ''); ?>">
    <input type="number" name="release_year" placeholder="Year" min="1900" max="2025" required
        value="<?php echo $edit_row['release_year'] ?? ''; ?>">
    <select name="genre" required>
        <option value="">-- Genre --</option>
        <?php foreach ($genres_list as $g): ?>
            <option <?php if (($edit_row['genre'] ?? '') === $g) echo 'selected'; ?>
                <?php echo $g; ?>
            </option>
        <?php endforeach; ?>
    </select>
    <button type="submit"><?php echo $edit_row ? 'Update' : 'Add'; ?></button>
    <?php if ($edit_row) echo '<a href="movies.php">Cancel</a>'; ?>
</form>

<form method="GET" class="inline">
    <label>Filter by genre:</label>
    <select name="filter" onchange="this.form.submit()">
        <option value="">All</option>
        <?php foreach ($genres_list as $g): ?>
            <option <?php if ($filter === $g) echo 'selected'; ?><?php echo $g; ?></option>
        <?php endforeach; ?>
    </select>
</form>

<table>

```



```

<tr><th>Title</th><th>Director</th><th>Year</th><th>Genre</th><th>Actions</th></tr>
<?php while ($r = $rows->fetch_assoc()): ?>
<tr>
    <td><?php echo htmlspecialchars($r['title']);    ?></td>
    <td><?php echo htmlspecialchars($r['director']); ?></td>
    <td><?php echo $r['release_year']; ?></td>
    <td><?php echo htmlspecialchars($r['genre']);    ?></td>
    <td>
        <a href="?edit=<?php echo $r['id']; ?>">Edit</a> |
        <a href="?delete=<?php echo $r['id']; ?>"
            onclick="return confirm('Delete?');">Delete</a>
    </td>
</tr>
<?php endwhile; ?>
</table>
</body></html>
<?php $conn->close(); ?>

```

Execution Instructions

Place this file inside C:\xampp\htdocs\<folder>\ (Windows) or /Applications/XAMPP/htdocs/<folder>/ (Mac). Start Apache and MySQL from the XAMPP Control Panel. Open phpMyAdmin (<http://localhost/phpmyadmin>) and run the SQL shown at the top of the file to create the database and table. Then visit <http://localhost/<folder>/<filename>> in your browser.

Q18. Hospital Patient Registration

XAMPP REQUIRED

Problem Statement

Design a patient registration system. Staff can register patients (Patient ID, Name, Age, Gender, Address). Validate inputs (Age must be numeric, Gender must be Male/Female/Other). Prevent duplicate Patient IDs. Display all registered patients in a searchable table.

File(s)

patients.php

Final Working Code

```
-- =====
-- SQL
-- =====
CREATE DATABASE IF NOT EXISTS qbank;
USE qbank;
CREATE TABLE IF NOT EXISTS patients (
    patient_id VARCHAR(20) PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    age INT NOT NULL,
    gender ENUM('Male', 'Female', 'Other') NOT NULL,
    address VARCHAR(255) NOT NULL
);

<?php
// ===== FILE: patients.php =====
$conn = new mysqli('localhost', 'root', '', 'qbank');
if ($conn->connect_error) die('DB error: ' . $conn->connect_error);

$msg = '';
$genders = ['Male', 'Female', 'Other'];

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $pid = trim($_POST['patient_id']);
    $name = trim($_POST['name']);
    $age = $_POST['age'];
    $gender = $_POST['gender'];
    $addr = trim($_POST['address']);

    if ($pid === '' || $name === '' || $addr === '') {
        $msg = '<div class="err">All fields are required.</div>';
    } elseif (!is_numeric($age) || $age <= 0 || $age > 150) {
        $msg = '<div class="err">Age must be a positive number.</div>';
    } elseif (!in_array($gender, $genders)) {
        $msg = '<div class="err">Gender must be Male, Female, or Other.</div>';
    } else {
        $chk = $conn->prepare('SELECT patient_id FROM patients WHERE patient_id = ?');
        $chk->bind_param('s', $pid);
        $chk->execute();
        $chk->store_result();
        if ($chk->num_rows > 0) {
            $msg = '<div class="err">Patient ID already exists.</div>';
        } else {
            $ins = $conn->prepare('INSERT INTO patients VALUES (?, ?, ?, ?, ?)');
            $ins->bind_param('ssiss', $pid, $name, $age, $gender, $addr);
            $ins->execute();
            $msg = '<div class="ok">Patient registered.</div>';
        }
    }
}

$search = trim($_GET['q'] ?? '');
if ($search !== '') {
    $like = '%' . $search . '%';
    $stmt = $conn->prepare(
        'SELECT * FROM patients WHERE patient_id LIKE ? OR name LIKE ? ORDER BY name'
    );
    $stmt->bind_param('ss', $like, $like);
```

```

$stmt->execute();
$rows = $stmt->get_result();
} else {
    $rows = $conn->query('SELECT * FROM patients ORDER BY name');
}
?>
<!DOCTYPE html>
<html><head><meta charset="UTF-8"><title>Patients</title>
<style>
    body{font-family:Arial;max-width:950px;margin:30px auto;padding:0 20px;}
    h2,h3{color:#333;}
    form.inline{display:flex;gap:8px;flex-wrap:wrap;align-items:center;
        background:#f4f4f4;padding:15px;border-radius:6px;margin-bottom:15px;}
    input,select{padding:8px;border:1px solid #ccc;border-radius:4px;}
    button{padding:8px 14px;background:#4CAF50;color:#fff;border:none;border-radius:4px;cursor:pointer;}
    table{width:100%;border-collapse:collapse;}
    th,td{border:1px solid #ddd;padding:10px;text-align:left;}
    th{background:#2196F3;color:#fff;}
    tr:nth-child(even){background:#f9f9f9;}
    .err{background:#f8d7da;color:#721c24;padding:10px;border-radius:4px;margin:10px 0;}
    .ok{background:#d4edda;color:#155724;padding:10px;border-radius:4px;margin:10px 0;}
</style></head><body>
<h2>Hospital Patient Registration</h2>
<?php echo $msg; ?>

<h3>Register New Patient</h3>
<form method="POST" class="inline">
    <input type="text" name="patient_id" placeholder="Patient ID" required>
    <input type="text" name="name" placeholder="Name" required>
    <input type="number" name="age" placeholder="Age" min="1" max="150" required>
    <select name="gender" required>
        <option value="">-- Gender --</option>
        <?php foreach ($genders as $g): ?>
            <option><?php echo $g; ?></option>
        <?php endforeach; ?>
    </select>
    <input type="text" name="address" placeholder="Address" required>
    <button type="submit">Register</button>
</form>

<form method="GET" class="inline">
    <input type="text" name="q" placeholder="Search by ID or Name"
        value="<?php echo htmlspecialchars($search); ?>">
    <button type="submit">Search</button>
    <?php if ($search) echo '<a href="patients.php">Clear</a>'; ?>
</form>

<table>
    <tr><th>ID</th><th>Name</th><th>Age</th><th>Gender</th><th>Address</th></tr>
    <?php while ($r = $rows->fetch_assoc()): ?>
    <tr>
        <td><?php echo htmlspecialchars($r['patient_id']); ?></td>
        <td><?php echo htmlspecialchars($r['name']); ?></td>
        <td><?php echo $r['age']; ?></td>
        <td><?php echo $r['gender']; ?></td>
        <td><?php echo htmlspecialchars($r['address']); ?></td>
    </tr>
    <?php endwhile; ?>
</table>
</body></html>
<?php $conn->close(); ?>

```

Execution Instructions

Place this file inside C:\xampp\htdocs\<folder> (Windows) or /Applications/XAMPP/htdocs/<folder> (Mac). Start Apache and MySQL from the XAMPP Control Panel. Open phpMyAdmin (<http://localhost/phpmyadmin>) and run the SQL shown at the top of the file to create the database and table. Then visit <http://localhost/<folder>/<filename>> in your browser.

Q19. Online Quiz Application

XAMPP REQUIRED

Problem Statement

Create a quiz system. Admin can add multiple-choice questions (Question, Option 1, Option 2, Option 3, Correct Answer). Validate that all options and the correct answer are provided. Users can take the quiz and their responses are stored in MySQL. Display the user's score at the end.

File(s)

quiz_admin.php, quiz.php, result.php

Final Working Code

```
-- =====
-- SQL
-- =====
CREATE DATABASE IF NOT EXISTS qbank;
USE qbank;
CREATE TABLE IF NOT EXISTS quiz_questions (
    id INT AUTO_INCREMENT PRIMARY KEY,
    question TEXT NOT NULL,
    option1 VARCHAR(255) NOT NULL,
    option2 VARCHAR(255) NOT NULL,
    option3 VARCHAR(255) NOT NULL,
    correct_answer ENUM('1','2','3') NOT NULL
);
CREATE TABLE IF NOT EXISTS quiz_responses (
    id INT AUTO_INCREMENT PRIMARY KEY,
    user_name VARCHAR(100) NOT NULL,
    score INT NOT NULL,
    total INT NOT NULL,
    taken_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

<?php
// ===== FILE: quiz_admin.php =====
$conn = new mysqli('localhost', 'root', '', 'qbank');
if ($conn->connect_error) die('DB error: ' . $conn->connect_error);
$msg = '';

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $q = trim($_POST['question']);
    $o1 = trim($_POST['option1']);
    $o2 = trim($_POST['option2']);
    $o3 = trim($_POST['option3']);
    $ca = $_POST['correct'];
    if ($q === '' || $o1 === '' || $o2 === '' || $o3 === '' || !in_array($ca, ['1','2','3'])) {
        $msg = '<div class="err">All fields are required and correct answer must be 1, 2, or 3.</div>';
    } else {
        $stmt = $conn->prepare(
            'INSERT INTO quiz_questions (question, option1, option2, option3, correct_answer)
            VALUES (?, ?, ?, ?, ?)'
        );
        $stmt->bind_param('sssss', $q, $o1, $o2, $o3, $ca);
        $stmt->execute();
        $msg = '<div class="ok">Question added.</div>';
    }
}

$total = $conn->query('SELECT COUNT(*) AS c FROM quiz_questions')->fetch_assoc()['c'];
?>
<!DOCTYPE html>
<html><head><meta charset="UTF-8"><title>Quiz Admin</title>
<style>
body{font-family:Arial;max-width:600px;margin:30px auto;padding:0 20px;}
h2{color:#333;}
form{background:#f4f4f4;padding:15px;border-radius:6px;}
input,textarea,select{width:100%;padding:8px;margin:5px 0;
border:1px solid #ccc;border-radius:4px;box-sizing:border-box;}
button{padding:10px 20px;background:#4CAF50;color:#fff;border:none;
border-radius:4px;cursor:pointer;}
```

```

        .err{background:#f8d7da;color:#721c24;padding:10px;border-radius:4px;margin:10px 0;}
        .ok{background:#d4edda;color:#155724;padding:10px;border-radius:4px;margin:10px 0;}
</style></head><body>
<h2>Quiz Admin &mdash; Add Question</h2>
<?php echo $msg; ?>
<p>Currently <?php echo $total; ?> questions in DB. <a href="quiz.php">Take the quiz</a></p>
<form method="POST">
    <textarea name="question" rows="3" placeholder="Question text" required></textarea>
    <input type="text" name="option1" placeholder="Option 1" required>
    <input type="text" name="option2" placeholder="Option 2" required>
    <input type="text" name="option3" placeholder="Option 3" required>
    <select name="correct" required>
        <option value="">-- Correct answer --</option>
        <option value="1">Option 1</option>
        <option value="2">Option 2</option>
        <option value="3">Option 3</option>
    </select>
    <button type="submit">Add Question</button>
</form>
</body></html>

<?php
// ===== FILE: quiz.php =====
$conn = new mysqli('localhost', 'root', '', 'qbank');
if ($conn->connect_error) die('DB error: ' . $conn->connect_error);

$rows = $conn->query('SELECT * FROM quiz_questions ORDER BY id');
?>
<!DOCTYPE html>
<html><head><meta charset="UTF-8"><title>Take Quiz</title>
<style>
    body{font-family:Arial;max-width:700px;margin:30px auto;padding:0 20px;}
    h2{color:#333;}
    .q{background:#f4f4f4;padding:15px;margin:12px 0;border-radius:6px;}
    label{display:block;margin:6px 0;}
    input[type=text]{width:100%;padding:10px;margin:8px 0;
        border:1px solid #ccc;border-radius:4px;box-sizing:border-box;}
    button{padding:12px 24px;background:#4CAF50;color:#fff;border:none;
        border-radius:4px;cursor:pointer;font-size:15px;}
</style></head><body>
<h2>Online Quiz</h2>
<form method="POST" action="result.php">
    <input type="text" name="user_name" placeholder="Your name" required>
    <?php while ($q = $rows->fetch_assoc()): ?>
        <div class="q">
            <strong>Q<?php echo $q['id']; ?>. <?php echo htmlspecialchars($q['question']); ?></strong>
            <label><input type="radio" name="ans[<?php echo $q['id']; ?>]" value="1" required>
                <?php echo htmlspecialchars($q['option1']); ?></label>
            <label><input type="radio" name="ans[<?php echo $q['id']; ?>]" value="2">
                <?php echo htmlspecialchars($q['option2']); ?></label>
            <label><input type="radio" name="ans[<?php echo $q['id']; ?>]" value="3">
                <?php echo htmlspecialchars($q['option3']); ?></label>
        </div>
    <?php endwhile; ?>
    <button type="submit">Submit Answers</button>
</form>
</body></html>

<?php
// ===== FILE: result.php =====
$conn = new mysqli('localhost', 'root', '', 'qbank');
if ($conn->connect_error) die('DB error: ' . $conn->connect_error);

if ($_SERVER['REQUEST_METHOD'] !== 'POST') { header('Location: quiz.php'); exit; }

$user = trim($_POST['user_name'] ?? 'Anonymous');
$ans = $_POST['ans'] ?? [];

$rows = $conn->query('SELECT id, correct_answer FROM quiz_questions');
$score = 0;
$total = 0;

```

```

while ($r = $rows->fetch_assoc()) {
    $total++;
    if (isset($ans[$r['id']]) && $ans[$r['id']] === $r['correct_answer']) $score++;
}

$stmt = $conn->prepare(
    'INSERT INTO quiz_responses (user_name, score, total) VALUES (?, ?, ?)'
);
$stmt->bind_param('sii', $user, $score, $total);
$stmt->execute();
?>
<!DOCTYPE html>
<html><head><meta charset="UTF-8"><title>Result</title>
<style>
    body{font-family:Arial;text-align:center;margin-top:80px;}
    h1{color:#4CAF50;}
    .score{font-size:48px;color:#2196F3;margin:20px 0;}
</style></head><body>
<h1>Quiz Complete, <?php echo htmlspecialchars($user); ?></h1>
<div class="score"><?php echo $score; ?> / <?php echo $total; ?></div>
<p>You got <?php echo $score; ?> out of <?php echo $total; ?> correct.</p>
<a href="quiz.php">Take again</a>
</body></html>

```

Execution Instructions

Place this file inside C:\xampp\htdocs\<folder>\ (Windows) or /Applications/XAMPP/htdocs/<folder>/ (Mac). Start Apache and MySQL from the XAMPP Control Panel. Open phpMyAdmin (<http://localhost/phpmyadmin>) and run the SQL shown at the top of the file to create the database and table. Then visit <http://localhost/<folder>/<filename>> in your browser. Three files are provided below: quiz_admin.php (add questions), quiz.php (take the quiz), and result.php (see score). Place all three in the same folder.

Set 2

AJAX, Vanilla JavaScript & jQuery Animations — Q20 to Q29

Q20. Weather App with AJAX (Vanilla JS)

BROWSER + API

Problem Statement

Create a weather application that fetches real-time weather data from OpenWeatherMap and displays it dynamically. Use XMLHttpRequest to fetch by city name. Show temperature, humidity, condition, and wind speed. Implement user-friendly error handling. Prevent page reload during data fetch.

File(s)

weather.html

Final Working Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Weather App</title>
<style>
  body { font-family: Arial; max-width: 450px; margin: 50px auto; padding: 20px;
    background: linear-gradient(135deg, #74b9ff, #a29bfe); color: #fff;
    border-radius: 10px; text-align: center; }
  h2 { margin-top: 0; }
  input { padding: 10px; width: 60%; border: none; border-radius: 4px;
    font-size: 14px; color: #333; }
  button { padding: 10px 18px; background: #2d3436; color: white;
    border: none; border-radius: 4px; cursor: pointer; }
  #result { margin-top: 20px; background: rgba(255,255,255,0.15);
    padding: 20px; border-radius: 8px; min-height: 80px; }
  .row { display: flex; justify-content: space-between; padding: 6px 0; }
  .err { color: #ffe66d; font-weight: bold; }
</style>
</head>
<body>
  <h2>Weather Now</h2>
  <input type="text" id="city" placeholder="Enter city name">
  <button onclick="getWeather()">Search</button>
  <div id="result">Enter a city to see the weather.</div>

<script>
// PASTE YOUR FREE OpenWeatherMap API KEY HERE:
const API_KEY = 'YOUR_API_KEY_HERE';

function getWeather() {
  const city = document.getElementById('city').value.trim();
  const result = document.getElementById('result');
  if (city === '') {
    result.innerHTML = '<span class="err">Please enter a city name.</span>';
    return;
  }
  result.innerHTML = 'Loading...';

  const xhr = new XMLHttpRequest();
  const url = 'https://api.openweathermap.org/data/2.5/weather?q=' +
    encodeURIComponent(city) + '&appid=' + API_KEY + '&units=metric';
  xhr.open('GET', url, true);

  xhr.onload = function () {
    if (xhr.status === 200) {
      const d = JSON.parse(xhr.responseText);
      result.innerHTML =
        '<h3>' + d.name + ', ' + d.sys.country + '</h3>' +
        '<div class="row"><span>Temperature:</span><span>' + d.main.temp + ' &deg;C</span></div>' +
        '<div class="row"><span>Condition:</span><span>' + d.weather[0].main + '</span></div>' +
        '<div class="row"><span>Humidity:</span><span>' + d.main.humidity + '%</span></div>' +
        '<div class="row"><span>Wind Speed:</span><span>' + d.wind.speed + ' m/s</span></div>';
    } else if (xhr.status === 404) {
      result.innerHTML = '<span class="err">City not found. Please check the spelling.</span>';
    } else if (xhr.status === 401) {
      result.innerHTML = '<span class="err">Invalid API key. Make sure your key is active.</span>';
    }
  }
}
```



```

    } else {
        result.innerHTML = '<span class="err">Error ' + xhr.status + '. Try again later.</span>';
    }
};

xhr.onerror = function () {
    result.innerHTML = '<span class="err">Network error. Check your internet connection.</span>';
};

xhr.send();
}

// Allow Enter key to trigger search
document.getElementById('city').addEventListener('keypress', function (e) {
    if (e.key === 'Enter') getWeather();
});
</script>
</body>
</html>

```

Execution Instructions

Save as weather.html and open in any browser. You need a free API key from <https://openweathermap.org/api> -- sign up, copy your key, and paste it into the API_KEY constant near the top of the script. New API keys can take a few minutes to activate.

Q21. Live User Search (Vanilla JS + AJAX)

BROWSER + API

Problem Statement

Build a search system that dynamically fetches and displays user data as the user types. Use the `oninput` event to trigger AJAX requests to JSONPlaceholder. Display results in a dropdown without refreshing the page. Show a loading indicator while fetching. Handle API errors gracefully.

File(s)

`live_user_search.html`

Final Working Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Live User Search</title>
<style>
  body { font-family: Arial; max-width: 500px; margin: 60px auto; padding: 0 20px; }
  h2 { color: #333; }
  #search { width: 100%; padding: 12px; box-sizing: border-box;
    border: 1px solid #ccc; border-radius: 4px; font-size: 14px; }
  #dropdown { border: 1px solid #ddd; border-top: none;
    max-height: 250px; overflow-y: auto; display: none; }
  .item { padding: 10px; border-bottom: 1px solid #eee; cursor: pointer; }
  .item:hover { background: #f0f8ff; }
  .item small { color: #777; }
  #loading { display: none; padding: 10px; color: #666; }
  .err { color: #c0392b; padding: 10px; }
</style>
</head>
<body>
  <h2>Live User Search</h2>
  <input type="text" id="search" placeholder="Type a user's name...">
  <div id="loading">Loading...</div>
  <div id="dropdown"></div>

<script>
const search = document.getElementById('search');
const drop  = document.getElementById('dropdown');
const load  = document.getElementById('loading');

let timer;
search.addEventListener('input', function () {
  clearTimeout(timer);
  const q = search.value.trim().toLowerCase();
  if (q === '') {
    drop.style.display = 'none';
    drop.innerHTML = '';
    return;
  }
  // Debounce so we only fire after 250ms idle
  timer = setTimeout(function () { fetchUsers(q); }, 250);
});

function fetchUsers(query) {
  load.style.display = 'block';
  drop.style.display = 'none';

  const xhr = new XMLHttpRequest();
  xhr.open('GET', 'https://jsonplaceholder.typicode.com/users', true);

  xhr.onload = function () {
    load.style.display = 'none';
    if (xhr.status === 200) {
      const all = JSON.parse(xhr.responseText);
      const matches = all.filter(function (u) {
        return u.name.toLowerCase().indexOf(query) !== -1 ||
          u.username.toLowerCase().indexOf(query) !== -1;
      });
    }
  };
}
```

```

    });
    renderResults(matches);
  } else {
    drop.innerHTML = '<div class="err">Error ' + xhr.status + '. Please try again.</div>';
    drop.style.display = 'block';
  }
};

xhr.onerror = function () {
  load.style.display = 'none';
  drop.innerHTML = '<div class="err">Network error. Check your connection.</div>';
  drop.style.display = 'block';
};

xhr.send();
}

function renderResults(matches) {
  if (matches.length === 0) {
    drop.innerHTML = '<div class="item"><em>No users found</em></div>';
  } else {
    drop.innerHTML = matches.map(function (u) {
      return '<div class="item"><strong>' + u.name + '</strong>' +
        '<br><small>@' + u.username + ' &middot; ' + u.email + '</small></div>';
    }).join('');
  }
  drop.style.display = 'block';
}
</script>
</body>
</html>

```

Execution Instructions

Save as live_user_search.html and open in any browser (internet required - uses public JSONPlaceholder API).

Q22. Dynamic To-Do List (AJAX + DOM Manipulation)

BROWSER + API

Problem Statement

Develop a to-do list that fetches tasks from JSONPlaceholder and allows users to add or delete tasks via AJAX without page reload. Show a spinner during API calls. Update the UI dynamically after each operation.

File(s)

ajax_todo.html

Final Working Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>AJAX To-Do List</title>
<style>
  body { font-family: Arial; max-width: 550px; margin: 40px auto; padding: 0 20px; }
  h2 { color: #333; }
  .input-row { display: flex; gap: 8px; margin-bottom: 15px; }
  #newTask { flex: 1; padding: 10px; border: 1px solid #ccc; border-radius: 4px; }
  button { padding: 10px 16px; background: #2196F3; color: #fff;
    border: none; border-radius: 4px; cursor: pointer; }
  button:hover { background: #1976D2; }
  .task { display: flex; justify-content: space-between; align-items: center;
    padding: 10px; margin: 5px 0; background: #f4f4f4; border-radius: 4px; }
  .del { background: #e74c3c; padding: 6px 12px; }
  #spinner { display: none; text-align: center; padding: 20px;
    font-size: 18px; color: #2196F3; }
  .spin { display: inline-block; width: 16px; height: 16px; border: 3px solid #ddd;
    border-top-color: #2196F3; border-radius: 50%;
    animation: spin 0.8s linear infinite; vertical-align: middle; }
  @keyframes spin { to { transform: rotate(360deg); } }
</style>
</head>
<body>
  <h2>AJAX To-Do List</h2>
  <div class="input-row">
    <input type="text" id="newTask" placeholder="Add a new task...">
    <button onClick="addTask()">Add</button>
  </div>
  <div id="spinner"><span class="spin"></span> Loading...</div>
  <div id="taskList"></div>

<script>
const API = 'https://jsonplaceholder.typicode.com/todos';
const list = document.getElementById('taskList');
const spinner = document.getElementById('spinner');

function showSpinner(on) { spinner.style.display = on ? 'block' : 'none'; }

function loadTasks() {
  showSpinner(true);
  fetch(API + '?_limit=5')
    .then(function (r) { return r.json(); })
    .then(function (data) {
      list.innerHTML = '';
      data.forEach(renderTask);
    })
    .catch(function () { list.innerHTML = '<p style="color:red">Failed to load tasks.</p>'; })
    .finally(function () { showSpinner(false); });
}

function renderTask(task) {
  const div = document.createElement('div');
  div.className = 'task';
  div.id = 'task-' + task.id;
```

```

div.innerHTML = '<span>' + task.title + '</span>' +
  '<button class="del" onclick="deleteTask(' + task.id + ')">Delete</button>';
list.appendChild(div);
}

function addTask() {
  const title = document.getElementById('newTask').value.trim();
  if (title === '') return;

  showSpinner(true);
  fetch(API, {
    method: 'POST',
    headers: { 'Content-Type': 'application/json' },
    body: JSON.stringify({ title: title, completed: false, userId: 1 })
  })
  .then(function (r) { return r.json(); })
  .then(function (newTask) {
    // JSONPlaceholder returns id 201 for any POST -- that's fine for the demo.
    // Make IDs unique within the session by using a timestamp.
    newTask.id = Date.now();
    renderTask(newTask);
    document.getElementById('newTask').value = '';
  })
  .catch(function () { alert('Failed to add task.');
})
  .finally(function () { showSpinner(false); });
}

function deleteTask(id) {
  showSpinner(true);
  fetch(API + '/' + id, { method: 'DELETE' })
  .then(function () {
    const el = document.getElementById('task-' + id);
    if (el) el.remove();
  })
  .catch(function () { alert('Failed to delete task.');
})
  .finally(function () { showSpinner(false); });
}

loadTasks();
</script>
</body>
</html>

```

Execution Instructions

Save as `ajax_todo.html` and open in any browser (internet required). Note: JSONPlaceholder is a fake REST API -- it accepts POST/DELETE requests and returns realistic responses, but data is not actually persisted on the server. The UI manages local state to simulate persistence within the session.

Q23. Country-State-City Dropdown (Event-Driven AJAX)

BROWSER

Problem Statement

Implement a cascading dropdown where selecting a country fetches states, and selecting a state fetches cities. Use the onchange event to trigger AJAX calls. Fetch from a mock JSON source. Populate dropdowns without page refresh and handle errors (e.g., no states found).

File(s)

cascade_dropdown.html

Final Working Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Cascading Dropdown</title>
<style>
  body { font-family: Arial; max-width: 500px; margin: 60px auto; padding: 0 20px; }
  h2 { color: #333; }
  label { display: block; margin: 10px 0 4px; font-weight: bold; }
  select { width: 100%; padding: 10px; border: 1px solid #ccc; border-radius: 4px;
    font-size: 14px; }
  select:disabled { background: #eee; cursor: not-allowed; }
  .status { margin-top: 4px; font-size: 12px; color: #666; }
  .err { color: #c0392b; }
</style>
</head>
<body>
  <h2>Country &arr; State &arr; City</h2>

  <label for="country">Country</label>
  <select id="country" onchange="loadStates()">
    <option value="">-- Select Country --</option>
    <option value="IN">India</option>
    <option value="US">United States</option>
    <option value="UK">United Kingdom</option>
  </select>

  <label for="state">State</label>
  <select id="state" disabled onchange="loadCities()">
    <option value="">-- Select State --</option>
  </select>
  <div class="status" id="stateStatus"></div>

  <label for="city">City</label>
  <select id="city" disabled>
    <option value="">-- Select City --</option>
  </select>
  <div class="status" id="cityStatus"></div>

<script>
// Mock data - in a real app this would be on the server, fetched via AJAX
const DATA = {
  IN: {
    'Maharashtra': ['Mumbai', 'Pune', 'Nagpur'],
    'Karnataka': ['Bangalore', 'Mysore'],
    'Tamil Nadu': ['Chennai', 'Coimbatore']
  },
  US: {
    'California': ['Los Angeles', 'San Francisco', 'San Diego'],
    'Texas': ['Houston', 'Austin', 'Dallas'],
    'New York': ['New York City', 'Buffalo']
  },
  UK: {
    'England': ['London', 'Manchester', 'Liverpool'],
    'Scotland': ['Edinburgh', 'Glasgow']
  }
}
```

```

    }
};

// Simulate an AJAX call. In production:
// fetch('/api/states?country=' + code).then(r => r.json()).then(callback)
function ajaxCall(payload, callback, errorCallback) {
    setTimeout(function () {
        if (payload) callback(payload);
        else errorCallback();
    }, 300);
}

function loadStates() {
    const code = document.getElementById('country').value;
    const stateSel = document.getElementById('state');
    const citySel = document.getElementById('city');
    const status = document.getElementById('stateStatus');

    // Reset dependent dropdowns
    stateSel.innerHTML = '<option value="">-- Loading... --</option>';
    stateSel.disabled = true;
    citySel.innerHTML = '<option value="">-- Select City --</option>';
    citySel.disabled = true;
    document.getElementById('cityStatus').textContent = '';

    if (code === '') {
        stateSel.innerHTML = '<option value="">-- Select State --</option>';
        status.textContent = '';
        return;
    }

    ajaxCall(DATA[code],
        function (states) {
            const keys = Object.keys(states);
            if (keys.length === 0) {
                stateSel.innerHTML = '<option value="">-- No states found --</option>';
                status.textContent = 'No states available for this country.';
                status.className = 'status err';
                return;
            }
            stateSel.innerHTML = '<option value="">-- Select State --</option>';
            keys.forEach(function (s) {
                stateSel.innerHTML += '<option value="' + s + '">' + s + '</option>';
            });
            stateSel.disabled = false;
            status.textContent = keys.length + ' state(s) loaded.';
            status.className = 'status';
        },
        function () {
            stateSel.innerHTML = '<option value="">-- Error --</option>';
            status.textContent = 'Failed to load states.';
            status.className = 'status err';
        }
    );
}

function loadCities() {
    const country = document.getElementById('country').value;
    const state = document.getElementById('state').value;
    const citySel = document.getElementById('city');
    const status = document.getElementById('cityStatus');

    citySel.innerHTML = '<option value="">-- Loading... --</option>';
    citySel.disabled = true;

    if (state === '') {
        citySel.innerHTML = '<option value="">-- Select City --</option>';
        status.textContent = '';
        return;
    }

    ajaxCall(DATA[country] && DATA[country][state],
        function (cities) {

```

```

    if (!cities || cities.length === 0) {
        citySel.innerHTML = '<option value="">-- No cities found --</option>';
        status.textContent = 'No cities available.';
        status.className = 'status err';
        return;
    }
    citySel.innerHTML = '<option value="">-- Select City --</option>';
    cities.forEach(function (c) {
        citySel.innerHTML += '<option value="' + c + '">' + c + '</option>';
    });
    citySel.disabled = false;
    status.textContent = cities.length + ' cities loaded.';
    status.className = 'status';
},
function () {
    citySel.innerHTML = '<option value="">-- Error --</option>';
    status.textContent = 'Failed to load cities.';
    status.className = 'status err';
}
);
</script>
</body>
</html>

```

Execution Instructions

Save as cascade_dropdown.html and open in any browser. The mock data is embedded as a JS object; the AJAX flow is simulated with setTimeout to mimic a real network call.

Q24. Login Form with AJAX (Form Handling + Validation)

BROWSER

Problem Statement

Create a login form that validates inputs and submits asynchronously. Validate email and password (non-empty, correct format). Submit data to a mock API via AJAX. Display success/error messages without page reload. Show a loading animation during submission.

File(s)

ajax_login.html

Final Working Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>AJAX Login</title>
<style>
  body { font-family: Arial; background: #eef2f7; }
  .card { max-width: 380px; margin: 60px auto; background: white;
    padding: 30px; border-radius: 8px; box-shadow: 0 2px 8px rgba(0,0,0,0.1); }
  h2 { text-align: center; color: #333; margin-top: 0; }
  input { width: 100%; padding: 10px; box-sizing: border-box;
    margin: 6px 0 12px; border: 1px solid #ccc; border-radius: 4px; }
  button { width: 100%; padding: 12px; background: #4CAF50; color: white;
    border: none; border-radius: 4px; cursor: pointer; font-size: 15px; }
  button:disabled { background: #aaa; cursor: not-allowed; }
  .message { margin-top: 10px; padding: 10px; border-radius: 4px; display: none; }
  .ok { background: #d4edda; color: #155724; }
  .err { background: #f8d7da; color: #721c24; }
  .spin { display: inline-block; width: 14px; height: 14px;
    border: 3px solid rgba(255,255,255,0.4); border-top-color: white;
    border-radius: 50%; animation: spin 0.8s linear infinite;
    vertical-align: middle; margin-right: 6px; }
  @keyframes spin { to { transform: rotate(360deg); } }
  .hint { font-size: 12px; color: #777; margin-top: 8px; text-align: center; }
</style>
</head>
<body>
  <div class="card">
    <h2>Login</h2>
    <input type="email" id="email" placeholder="Email">
    <input type="password" id="password" placeholder="Password">
    <button id="loginBtn" onclick="doLogin()">Sign In</button>
    <div id="message" class="message"></div>
    <p class="hint">Demo: email <strong>demo@test.com</strong>, any password 6+ chars.</p>
  </div>

  <script>
function showMsg(text, ok) {
  const m = document.getElementById('message');
  m.textContent = text;
  m.className = 'message ' + (ok ? 'ok' : 'err');
  m.style.display = 'block';
}

function doLogin() {
  const email = document.getElementById('email').value.trim();
  const pass = document.getElementById('password').value;
  const btn = document.getElementById('loginBtn');

  // Validation
  if (!/^[^\s@]+@[^\s@]+\.[^\s@]+$/ .test(email)) {
    showMsg('Please enter a valid email address.', false); return;
  }
  if (pass.length < 6) {
    showMsg('Password must be at least 6 characters.', false); return;
  }
}
```

```

}

// Show loading state
btn.disabled = true;
btn.innerHTML = '<span class="spin"></span> Signing in...';
document.getElementById('message').style.display = 'none';

// Simulate AJAX call. In production:
//   fetch('/api/login', { method:'POST', body: JSON.stringify({email,pass}) })
setTimeout(function () {
  btn.disabled = false;
  btn.textContent = 'Sign In';

  if (email === 'demo@test.com' && pass.length >= 6) {
    showMsg('Login successful! Welcome, ' + email, true);
  } else {
    showMsg('Invalid credentials. Please try again.', false);
  }
}, 1200);
}

// Allow Enter key
document.getElementById('password').addEventListener('keypress', function (e) {
  if (e.key === 'Enter') doLogin();
});
</script>
</body>
</html>

```

Execution Instructions

Save as `ajax_login.html` and open in any browser. The login is simulated locally - any password 6+ chars matched with email 'demo@test.com' succeeds (everything else fails).

Q25. Comment System (AJAX + DOM Updates)

BROWSER + API

Problem Statement

Build a comment section where users can post and view comments in real-time. Fetch existing comments from JSONPlaceholder. Allow new comments via AJAX. Validate comment input (non-empty). Display new comments with a fade-in animation.

File(s)

comments.html

Final Working Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Comment System</title>
<style>
  body { font-family: Arial; max-width: 600px; margin: 40px auto; padding: 0 20px; }
  h2 { color: #333; }
  textarea { width: 100%; padding: 10px; border: 1px solid #ccc;
    border-radius: 4px; box-sizing: border-box; resize: vertical; }
  input { width: 100%; padding: 10px; border: 1px solid #ccc;
    border-radius: 4px; box-sizing: border-box; margin-bottom: 8px; }
  button { padding: 10px 20px; background: #2196F3; color: white;
    border: none; border-radius: 4px; cursor: pointer; margin-top: 8px; }
  .comment {
    padding: 12px; margin: 10px 0; background: #f4f4f4;
    border-left: 4px solid #2196F3; border-radius: 4px;
    opacity: 0; transition: opacity 0.6s ease;
  }
  .comment.show { opacity: 1; }
  .comment .name { font-weight: bold; color: #1976D2; }
  .err { color: #c0392b; margin-top: 5px; }
</style>
</head>
<body>
  <h2>Comments</h2>

  <h3>Leave a comment</h3>
  <input type="text" id="name" placeholder="Your name">
  <textarea id="body" rows="3" placeholder="Your comment..."></textarea>
  <div id="error" class="err"></div>
  <button onclick="postComment()">Post Comment</button>

  <h3>Recent Comments</h3>
  <div id="commentList">Loading...</div>

<script>
const API = 'https://jsonplaceholder.typicode.com/comments';
const list = document.getElementById('commentList');

function loadComments() {
  fetch(API + '?_limit=5')
    .then(function (r) { return r.json(); })
    .then(function (data) {
      list.innerHTML = '';
      data.forEach(function (c) { addCommentToDOM(c.name, c.body); });
    })
    .catch(function () { list.innerHTML = '<p style="color:red">Failed to load comments.</p>'; });
}

function addCommentToDOM(name, body) {
  const div = document.createElement('div');
  div.className = 'comment';
  div.innerHTML = '<div class="name">' + escapeHtml(name) + '</div>' +
    '<div>' + escapeHtml(body) + '</div>';
  list.prepend(div);
  // Trigger fade-in on next frame
}
```

```

    requestAnimationFrame(function () {
      requestAnimationFrame(function () { div.classList.add('show'); });
    });
  }

function escapeHtml(s) {
  return s.replace(/[\<>"']/g, function (c) {
    return {'&': '&amp;', '<': '&lt;', '>': '&gt;', '"': '&quot;', "'": '&#39;'}[c];
  });
}

function postComment() {
  const name = document.getElementById('name').value.trim();
  const body = document.getElementById('body').value.trim();
  const err = document.getElementById('error');

  if (name === '' || body === '') {
    err.textContent = 'Both name and comment are required.';
    return;
  }
  err.textContent = '';

  fetch(API, {
    method: 'POST',
    headers: { 'Content-Type': 'application/json' },
    body: JSON.stringify({ name: name, body: body, postId: 1 })
  })
  .then(function (r) { return r.json(); })
  .then(function () {
    addCommentToDOM(name, body);
    document.getElementById('name').value = '';
    document.getElementById('body').value = '';
  })
  .catch(function () { err.textContent = 'Failed to post comment.'; });
}

loadComments();
</script>
</body>
</html>

```

Execution Instructions

Save as comments.html and open in any browser (internet required for JSONPlaceholder).

Q26. Animated News Feed (jQuery + AJAX)

BROWSER + API

Problem Statement

Fetch news headlines and display them with smooth animations. Use jQuery AJAX. Display items one by one using `slideDown()`. Show a loading spinner. Allow users to refresh news with an animation.

File(s)

news_feed.html

Final Working Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Animated News Feed</title>
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<style>
  body { font-family: Arial; max-width: 700px; margin: 30px auto; padding: 0 20px;
    background: #f5f5f5; }
  h2 { color: #333; }
  #refresh { padding: 10px 20px; background: #2196F3; color: white;
    border: none; border-radius: 4px; cursor: pointer; }
  #refresh:hover { background: #1976D2; }
  #spinner { display: none; text-align: center; padding: 30px; }
  .spin { display: inline-block; width: 30px; height: 30px;
    border: 4px solid #ddd; border-top-color: #2196F3;
    border-radius: 50%; animation: spin 0.8s linear infinite; }
  @keyframes spin { to { transform: rotate(360deg); } }
  .news-item {
    display: none; background: white; padding: 15px; margin: 10px 0;
    border-radius: 6px; box-shadow: 0 1px 3px rgba(0,0,0,0.1);
    border-left: 4px solid #2196F3;
  }
  .news-item h3 { margin: 0 0 8px; color: #1976D2; text-transform: capitalize; }
  .news-item p { margin: 0; color: #555; line-height: 1.5; }
</style>
</head>
<body>
  <h2>Latest News</h2>
  <button id="refresh">&#x21bb; Refresh News</button>

  <div id="spinner"><span class="spin"></span></div>
  <div id="newsFeed"></div>

<script>
$(document).ready(function () {
  function loadNews() {
    $('#newsFeed').empty();
    $('#spinner').show();

    $.ajax({
      url: 'https://jsonplaceholder.typicode.com/posts?_limit=6',
      method: 'GET',
      success: function (data) {
        $('#spinner').hide();
        // Animate items in one by one
        data.forEach(function (item, i) {
          var $div = $('<div class="news-item"></div>')
            .append('<h3>' + item.title + '</h3>')
            .append('<p>' + item.body + '</p>');
          $('#newsFeed').append($div);
          $div.delay(i * 200).slideDown(400);
        });
      },
      error: function () {
        $('#spinner').hide();
        $('#newsFeed').html('<p style="color:red">Failed to load news. Please try again.</p>');
      }
    });
  }
});
```

```

}

$('#refresh').click(function () {
    $('#refresh').prop('disabled', true).text('Refreshing...');
    $('#newsFeed').children().slideUp(300, function () { $(this).remove(); });
    setTimeout(function () {
        loadNews();
        $('#refresh').prop('disabled', false).html('&#x21bb; Refresh News');
    }, 350);
});

loadNews();
});
</script>
</body>
</html>

```

Execution Instructions

Save as news_feed.html and open in any browser. Uses JSONPlaceholder /posts as a news source so no API key is needed. To use the real NewsAPI, sign up at <https://newsapi.org> and replace the URL accordingly.

Q27. Shopping Cart (AJAX + jQuery Animations)

BROWSER + API

Problem Statement

Create an interactive shopping cart where users can add/remove items dynamically. Fetch product data from a mock API. Add items via AJAX POST. Remove items with a `fadeOut()` animation. Update cart total without page reload.

File(s)

ajax_cart.html

Final Working Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>AJAX Shopping Cart</title>
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<style>
body { font-family: Arial; max-width: 1000px; margin: 30px auto; padding: 0 20px; }
h2, h3 { color: #333; }
.layout { display: flex; gap: 20px; }
.products, .cart { flex: 1; background: #fafafa; padding: 15px; border-radius: 6px; }
.product, .cart-item {
padding: 10px; margin: 8px 0; background: #fff;
border-radius: 4px; border-left: 4px solid #2196F3;
display: flex; justify-content: space-between; align-items: center;
}
.cart-item { border-left-color: #4CAF50; }
button { padding: 6px 12px; background: #2196F3; color: white;
border: none; border-radius: 4px; cursor: pointer; }
.remove { background: #e74c3c; }
#total { margin-top: 15px; font-size: 18px; font-weight: bold; color: #4CAF50; }
</style>
</head>
<body>
<h2>Shopping Cart</h2>
<div class="layout">
<div class="products">
<h3>Products</h3>
<div id="productList">Loading...</div>
</div>
<div class="cart">
<h3>Your Cart</h3>
<div id="cartList"></div>
<div id="total">Total: $0.00</div>
</div>
</div>

<script>
$(document).ready(function () {
var total = 0;
var nextItemId = 1;

// Fetch product data from mock API. In a real shop you'd hit /api/products
// and get real product objects. JSONPlaceholder /photos gives us titles+ids.
$.get('https://jsonplaceholder.typicode.com/photos?_limit=6', function (data) {
$('#productList').empty();
data.forEach(function (p) {
// Make up a price based on id so it's deterministic
var price = parseFloat((10 + (p.id % 50)).toFixed(2));
var $item = $('<div class="product"></div>')
.append('<span><strong>' + p.title.substring(0, 30) + '</strong>'
+ '<br><small>$' + price + '</small></span>')
.append('<button>Add to Cart</button>')
.click(function () { addToCart(p.title.substring(0, 30), price); });
$('#productList').append($item);
});
}).fail(function () {
```

```

$('#productList').html('<p style="color:red">Failed to load products.</p>');
});

function addToCart(name, price) {
  // Simulate POST to /api/cart in a real app
  $.ajax({
    url: 'https://jsonplaceholder.typicode.com/posts',
    method: 'POST',
    data: JSON.stringify({ title: name, price: price }),
    contentType: 'application/json',
    success: function () {
      var id = 'cart-' + (nextItemId++);
      var $item = $('<div class="cart-item" id="' + id + '"></div>')
        .append('<span>' + name + ' &mdash; $' + price.toFixed(2) + '</span>')
        .append('<button class="remove">Remove</button>')
        .click(function () { removeFromCart(id, price); });
      $item.hide().appendTo('#cartList').fadeIn(300);
      total += price;
      updateTotal();
    },
    error: function () { alert('Failed to add item.'); }
  });
}

function removeFromCart(id, price) {
  $('#' + id).fadeOut(300, function () {
    $(this).remove();
    total -= price;
    updateTotal();
  });
}

function updateTotal() {
  $('#total').text('Total: $' + total.toFixed(2));
}
});
</script>
</body>
</html>

```

Execution Instructions

Save as ajax_cart.html and open in any browser (internet required for JSONPlaceholder).

Q28. Infinite Scroll (jQuery + AJAX)

BROWSER + API

Problem Statement

Implement an infinite scroll feature that loads more content as the user scrolls down. Fetch initial posts from JSONPlaceholder. Load more when the user reaches the bottom. Use `fadeIn()` for smooth content loading. Show a 'Loading...' indicator during fetch.

File(s)

infinite_scroll.html

Final Working Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Infinite Scroll</title>
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<style>
  body { font-family: Arial; max-width: 700px; margin: 0 auto; padding: 20px;
    background: #f5f5f5; }
  h2 { color: #333; text-align: center; }
  .post {
    display: none; background: white; padding: 15px; margin: 10px 0;
    border-radius: 6px; box-shadow: 0 1px 3px rgba(0,0,0,0.1);
  }
  .post h3 { margin: 0 0 8px; color: #1976D2; }
  .post p { margin: 0; color: #555; line-height: 1.5; }
  #loader { text-align: center; padding: 20px; font-weight: bold;
    color: #2196F3; display: none; }
  #end { text-align: center; padding: 20px; color: #777;
    font-style: italic; display: none; }
</style>
</head>
<body>
  <h2>Infinite Scroll Demo</h2>
  <div id="posts"></div>
  <div id="loader">Loading more posts...</div>
  <div id="end">You've reached the end!</div>

<script>
$(document).ready(function () {
  var page = 1;
  var perPage = 5;
  var loading = false;
  var done = false;

  function loadPosts() {
    if (loading || done) return;
    loading = true;
    $('#loader').show();

    $.ajax({
      url: 'https://jsonplaceholder.typicode.com/posts',
      data: { _page: page, _limit: perPage },
      success: function (data) {
        $('#loader').hide();
        loading = false;

        if (data.length === 0) {
          done = true;
          $('#end').show();
          return;
        }

        data.forEach(function (p) {
          var $div = $('<div class="post"></div>')
            .append('<h3>' + p.title + '</h3>')
            .append('<p>' + p.body + '</p>');
        });
      }
    });
  }
});

```

```

        $('#posts').append($div);
        $div.fadeIn(400);
    });

    page++;
    // JSONPlaceholder has 100 posts; stop after we've loaded them all
    if (page * perPage > 100) { done = true; $('#end').show(); }
},
error: function () {
    $('#loader').hide();
    loading = false;
    $('#posts').append('<p style="color:red;text-align:center;">Failed to load posts.</p>');
}
});
}

// Trigger load when user scrolls near the bottom
$(window).scroll(function () {
    if ($(window).scrollTop() + $(window).height() >= $(document).height() - 100) {
        loadPosts();
    }
});

loadPosts(); // Initial load
});
</script>
</body>
</html>

```

Execution Instructions

Save as infinite_scroll.html and open in any browser (internet required).

Q29. Interactive Poll System (AJAX + Real-Time Updates)

BROWSER

Problem Statement

Build a poll system where users can vote and see real-time results. Fetch poll questions and options. Allow users to submit votes via AJAX. Update vote counts dynamically with slideUp/slideDown. Prevent multiple votes from the same user.

File(s)

poll.html

Final Working Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Poll System</title>
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<style>
  body { font-family: Arial; max-width: 500px; margin: 40px auto; padding: 0 20px; }
  h2 { color: #333; }
  .question { font-size: 18px; margin-bottom: 15px; font-weight: bold; }
  .option { padding: 10px; margin: 8px 0; background: #f4f4f4;
    border-radius: 4px; cursor: pointer; transition: background 0.3s; }
  .option:hover { background: #e3f2fd; }
  #results { display: none; margin-top: 20px; }
  .result-bar { background: #ddd; border-radius: 4px; overflow: hidden; height: 30px;
    margin: 8px 0; position: relative; }
  .fill { background: linear-gradient(90deg, #4CAF50, #8BC34A); height: 100%;
    width: 0%; transition: width 0.6s ease; }
  .label { position: absolute; left: 10px; top: 6px; font-weight: bold; color: #333; }
  .pct { position: absolute; right: 10px; top: 6px; font-weight: bold; color: #333; }
  #voted { display: none; padding: 10px; background: #d4edda; color: #155724;
    border-radius: 4px; margin-top: 10px; }
  #reset { background: #e74c3c; color: white; border: none; padding: 8px 16px;
    border-radius: 4px; cursor: pointer; margin-top: 10px; }
</style>
</head>
<body>
  <h2>Quick Poll</h2>
  <div class="question" id="question"></div>
  <div id="optionsBox"></div>

  <div id="voted">Thanks for voting!</div>
  <div id="results"></div>
  <button id="reset" style="display:none;" onclick="resetVote()">Vote Again</button>

<script>
// Mock poll data. In production, this would come from the server.
var poll = {
  id: 'poll-1',
  question: "What's your favorite frontend framework?",
  options: [
    { id: 'react', text: 'React', votes: 47 },
    { id: 'vue', text: 'Vue', votes: 32 },
    { id: 'angular', text: 'Angular', votes: 18 },
    { id: 'svelte', text: 'Svelte', votes: 24 }
  ]
};

$(document).ready(function () {
  $('#question').text(poll.question);

  // Show options if user hasn't voted yet
  if (localStorage.getItem(poll.id)) {
    showResults();
  } else {
```

```

    poll.options.forEach(function (o) {
        $('<div class="option"></div>').text(o.text)
            .data('id', o.id).appendTo('#optionsBox');
    });
}

$(document).on('click', '.option', function () {
    var optId = $(this).data('id');
    submitVote(optId);
});

function submitVote(optionId) {
    // Simulate AJAX POST. In production:
    // $.post('/api/vote', { poll: poll.id, option: optionId }, ...)
    setTimeout(function () {
        // "Server" updates the vote count
        poll.options.forEach(function (o) {
            if (o.id === optionId) o.votes++;
        });
        localStorage.setItem(poll.id, optionId);

        // Smooth UI transition
        $('#optionsBox').slideUp(400, function () {
            $(this).empty();
            $('#voted').slideDown(300);
            showResults();
        });
    }, 400);
}

function showResults() {
    var total = poll.options.reduce(function (s, o) { return s + o.votes; }, 0);
    var html = '';
    poll.options.forEach(function (o) {
        var pct = total === 0 ? 0 : Math.round((o.votes / total) * 100);
        html += '<div class="result-bar">' +
            '<div class="fill" data-pct="' + pct + '"></div>' +
            '<span class="label">' + o.text + ' (' + o.votes + ')</span>' +
            '<span class="pct">' + pct + '%</span>' +
            '</div>';
    });
    $('#results').html(html).slideDown(400);
    // Animate the bars
    setTimeout(function () {
        $('.fill').each(function () { $(this).css('width', $(this).data('pct') + '%'); });
    }, 100);
    $('#reset').show();
}

function resetVote() {
    localStorage.removeItem(poll.id);
    location.reload();
}
</script>
</body>
</html>

```

Execution Instructions

Save as poll.html and open in any browser. Vote tracking uses localStorage to prevent multiple votes from the same browser. Clear the localStorage to vote again.